第十章　双口网络

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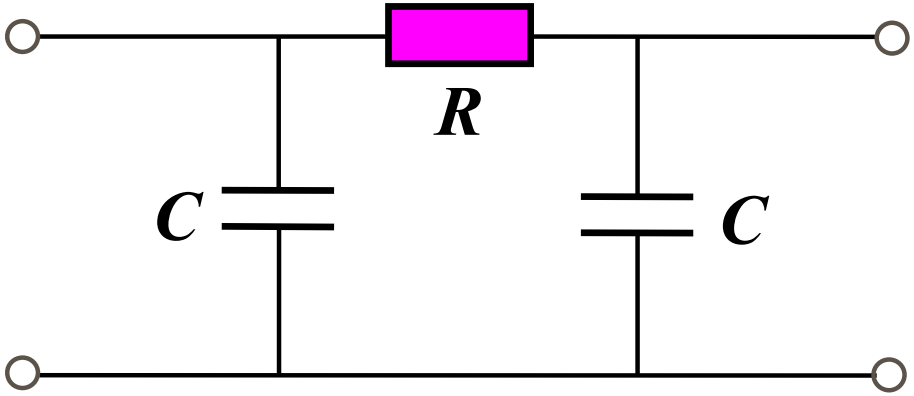
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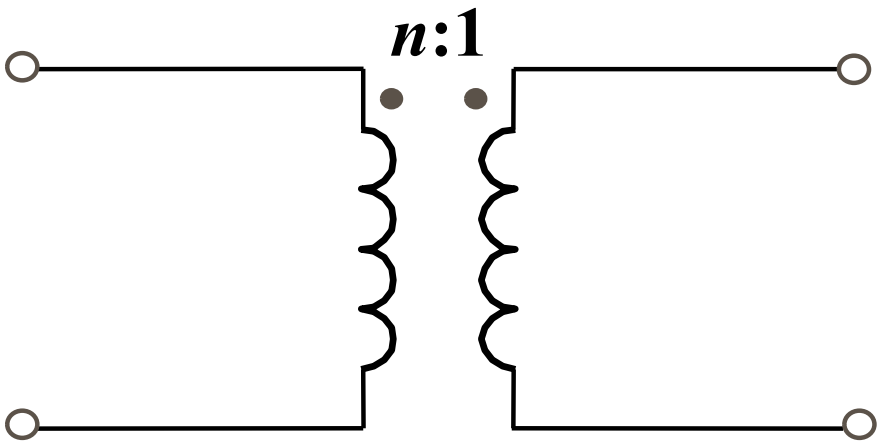
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10.1 双口网络概述

双口网络在工程中应用广泛，如互感器、变压器、晶体管放大器、滤波网络等。当不研究内部形状都属于双口网络。

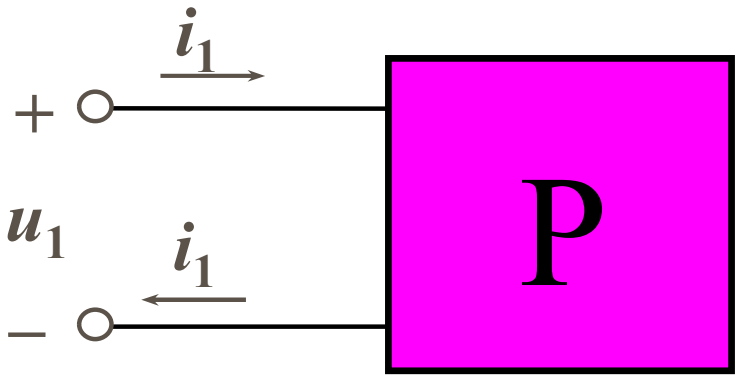


滤波器



变压器

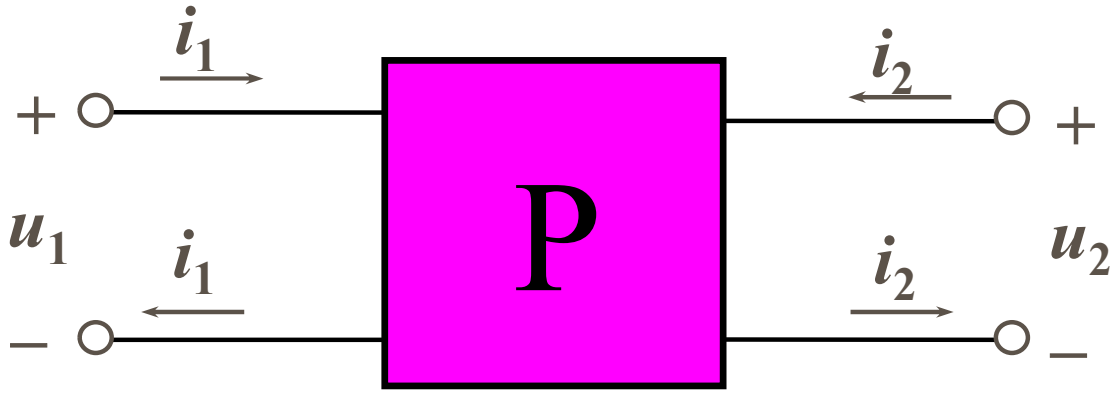
1） 一端口网络



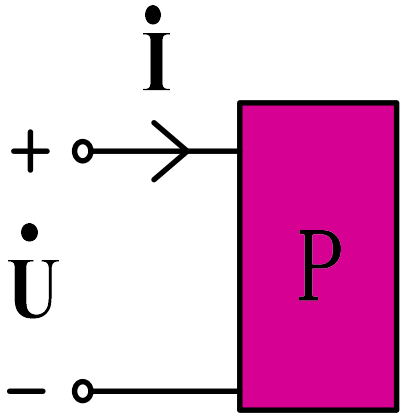
端口由一对端钮构成，从一个端钮流入的电流等于从另一个端钮流出的电流。

2） 双口（二端口)网络

对外具有两个端口的网络称为二端口网络，简称为双口网络。双口网络的每一个端口的电流必须是一进一出相等。



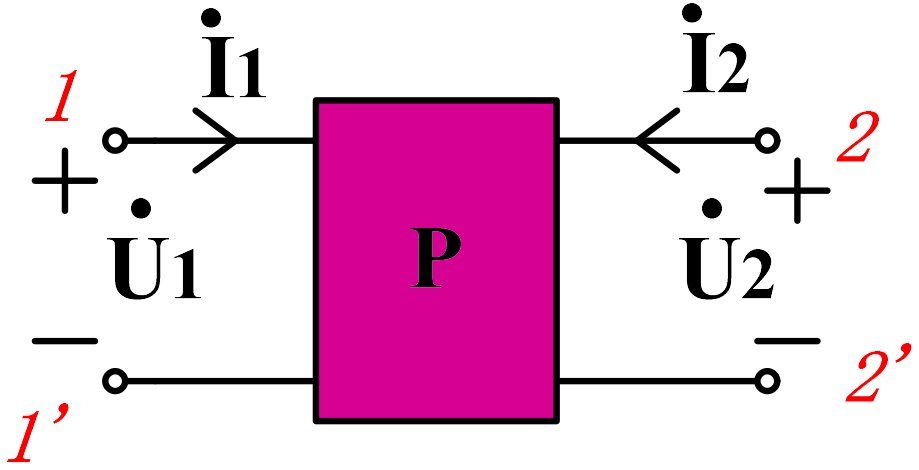
3）双口网络的参数



$$\dot{U} = Z\dot{I}$$

$$\dot{I} = Y\dot{U}$$

单端口网络端口电压电流之间的关系用一个参数表示。



双口网络有四个端部变量：

$\dot{U}_1$ $\dot{U}_2$ $\dot{I}_1$ $\dot{I}_2$

任意二个变量可表示为另二个变量的线性方程，反映端口电压和电流的关系。（如二个端口电压可表示为二个端口电流的线性方程）。

根据所选变量不同，一个双口网络共有6种特性参数：

短路参数Y；$\dot{U}_1,\dot{U}_2 \rightarrow \dot{I}_1,\dot{I}_2$

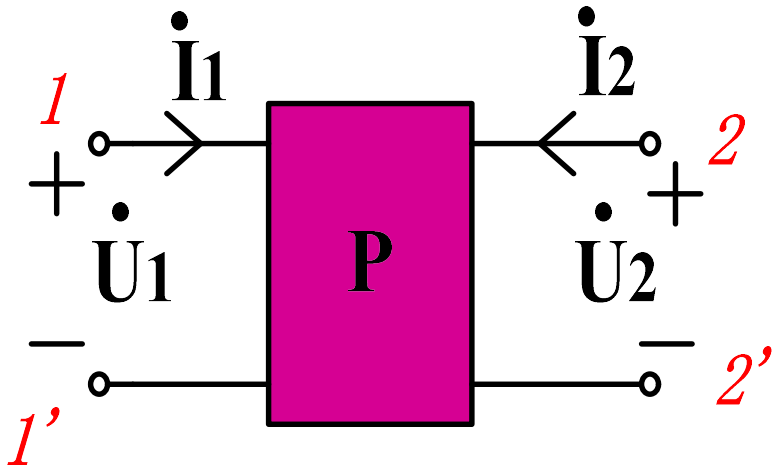
开路参数Z；$\dot{I}_1,\dot{I}_2 \rightarrow \dot{U}_1,\dot{U}_2$

传输参数T；$\dot{U}_2,\dot{I}_2 \rightarrow \dot{U}_1,\dot{I}_1$

逆传输参数T '；

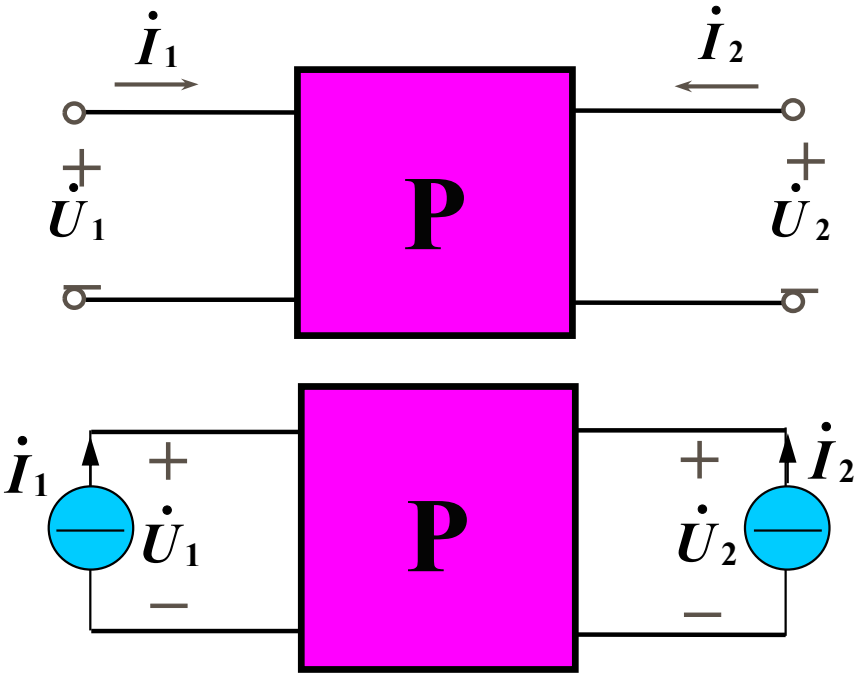
混合参数H；$\dot{I}_1,\dot{U}_2 \rightarrow \dot{U}_1,\dot{I}_2$

逆混合参数H '.



10.2 双口网络的开路参数Z

1、Z 参数和方程



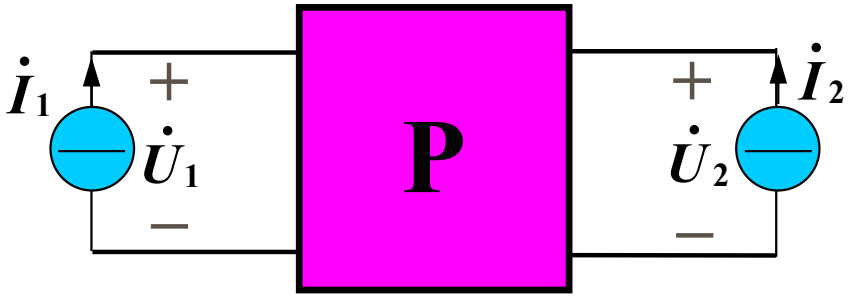
1）方程：

采用相量形式(正弦稳态)。将两个端口各施加一电流源，则端口电压可视为这些电流源的叠加作用产生。

即

$$\begin{cases} \dot{U}_1 = Z_{11}\dot{I}_1 + Z_{12}\dot{I}_2 \\ \dot{U}_2 = Z_{21}\dot{I}_1 + Z_{22}\dot{I}_2 \end{cases}$$

$$\begin{cases} \dot{U}_1 = Z_{11}\dot{I}_1 + Z_{12}\dot{I}_2 \\ \dot{U}_2 = Z_{21}\dot{I}_1 + Z_{22}\dot{I}_2 \end{cases}$$



上述方程即为*Z* 参数方程，其系数即为 *Z* 参数，

写成矩阵形式为：

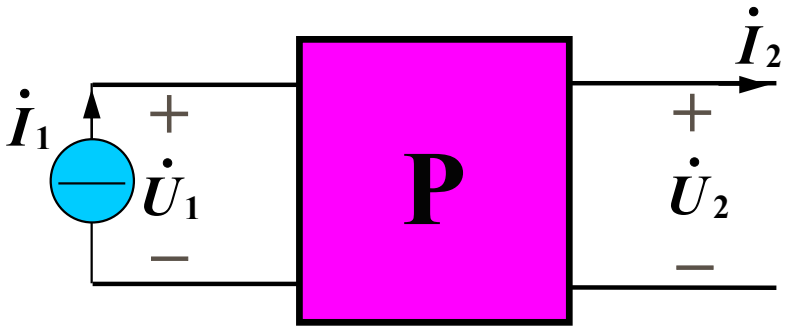
Z 参数矩阵

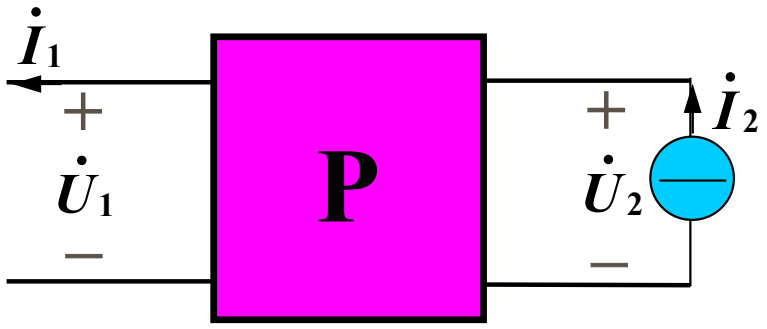
$$\begin{bmatrix} \dot{U}_1 \\ \dot{U}_2 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} = [Z]\dot{I} \qquad [Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}$$

Z参数反映了端部电流与端电压关系，若端电流已知，则由Z参数可求出端电压。

2）Z参数含义及计算：

$$\begin{cases} \dot{U}_1 = Z_{11}\dot{I}_1 + Z_{12}\dot{I}_2 \\ \dot{U}_2 = Z_{21}\dot{I}_1 + Z_{22}\dot{I}_2 \end{cases}$$





$$Z_{11} = \left.\frac{\dot{U}_1}{\dot{I}_1}\right|_{\dot{I}_2=0} \qquad Z_{12} = \left.\frac{\dot{U}_1}{\dot{I}_2}\right|_{\dot{I}_1=0}$$

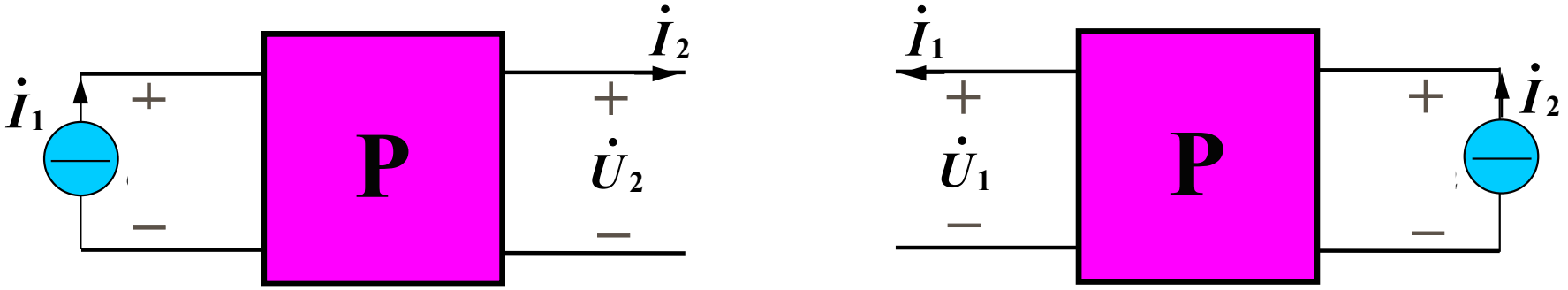
$$Z_{21} = \left.\frac{\dot{U}_2}{\dot{I}_1}\right|_{\dot{I}_2=0} \qquad Z_{22} = \left.\frac{\dot{U}_2}{\dot{I}_2}\right|_{\dot{I}_1=0}$$

它们都是开路情况下的阻抗，所以又称Z参数为开路参数。

3）互易双口网络(满足互易定理——无独立源无受控源网络）

满足 $Z_{12}=Z_{21}$

$$\mathbf{Z}_{12}=\left.\frac{\dot{\mathbf{U}}_1}{\dot{\mathbf{I}}_2}\right|_{\dot{\mathbf{I}}_1=0},\qquad \mathbf{Z}_{21}=\left.\frac{\dot{\mathbf{U}}_2}{\dot{\mathbf{I}}_1}\right|_{\dot{\mathbf{I}}_2=0}$$



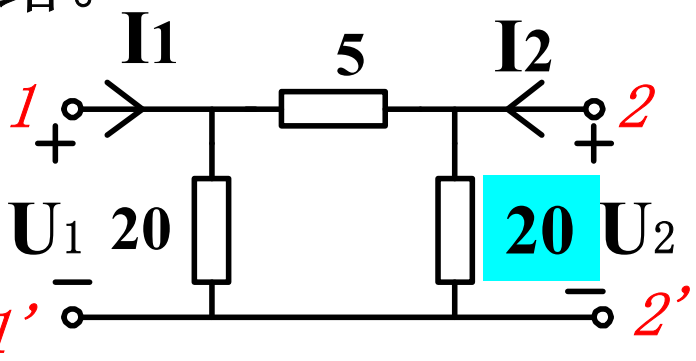
当 $\dot{I}_1=\dot{I}_2$时，$\dot{\mathbf{U}}_1=\dot{\mathbf{U}}_2$ $\therefore Z_{12}=Z_{21}$

互易双口网络四个参数中只有三个是独立的。

4） 对称双口网络

$Z_{12}=Z_{21}$，$Z_{11}=Z_{22}$，　　称为对称双口网络

电路结构左右对称的一般为对称双口网络。

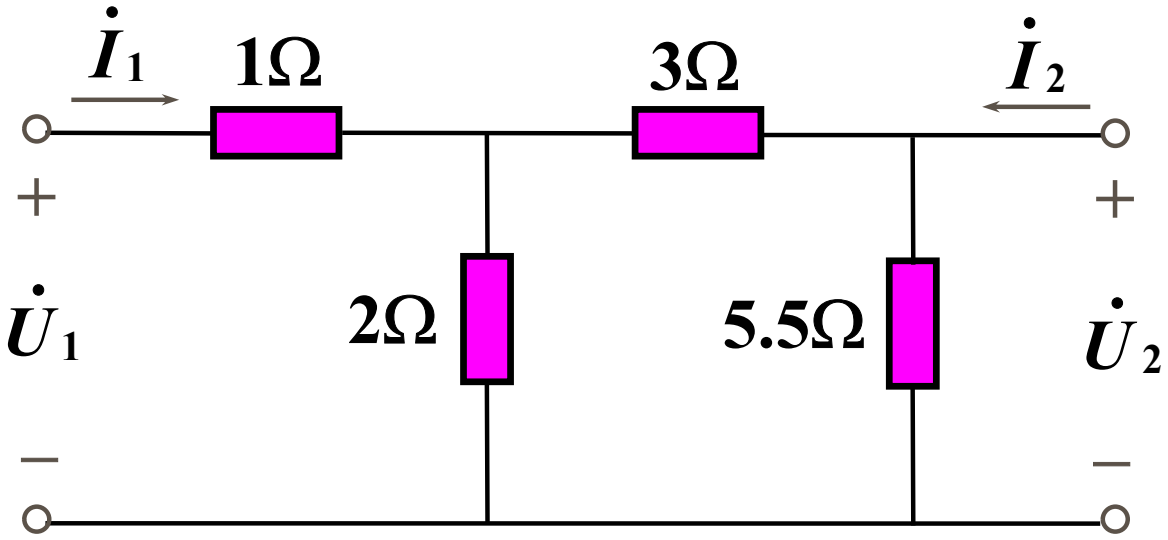


上例中，15→20时，则 $Z_{11}=Z_{22}$

对称双口网络只有两个参数是独立的。

对称双口网络是指两个端口电气特性上对称。结构不对称的二端口，其电气特性可能是对称的。这样的双口网络也是对称双口网络。

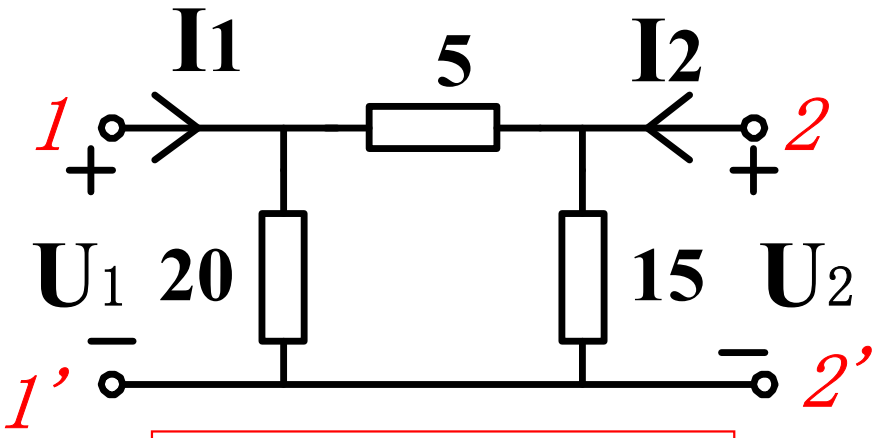
例



$$\begin{cases} Z_{11} = Z_{22} = 2.619\Omega \\ Z_{12} = Z_{21} = 1.0476\Omega \end{cases}$$

5）Z 参数的计算

例1　如图电路，求Z参数。



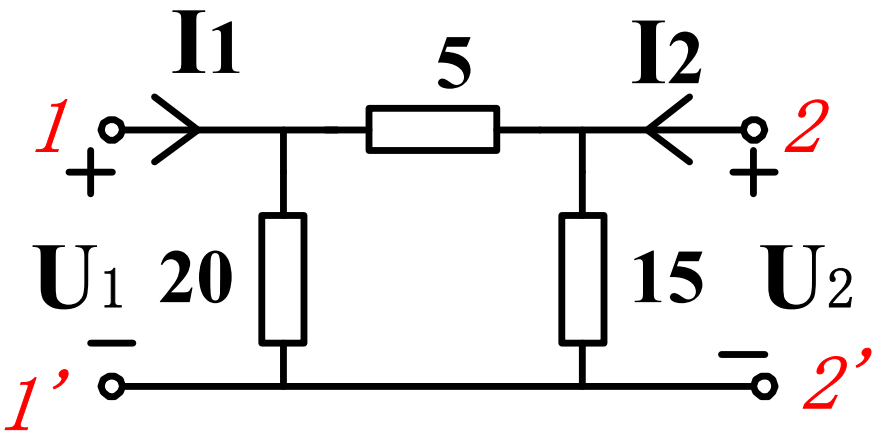
解：定义法：

由Z参数的定义，得：

$$Z_{11}=\left.\frac{\dot{U}_1}{\dot{I}_1}\right|_{i_2=0}=\frac{20\times(15+5)}{20+(15+5)}=10\quad\Omega$$

$$\begin{cases}\dot{U}_1=Z_{11}\dot{I}_1+Z_{12}\dot{I}_2\\ \dot{U}_2=Z_{21}\dot{I}_1+Z_{22}\dot{I}_2\end{cases}$$

$$Z_{21}=\left.\frac{\dot{U}_2}{\dot{I}_1}\right|_{\dot{I}_2=0}=\frac{\frac{1}{2}\dot{I}_1\times 15}{\dot{I}_1}=7.5\quad\Omega$$



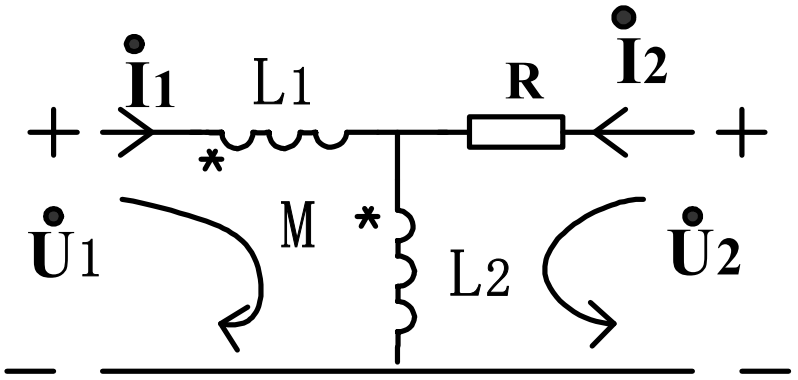
$$\mathbf{Z_{12}} = \left.\frac{\dot{U}_1}{\dot{I}_2}\right|_{\dot{I}_1=0} = \frac{20\times\dfrac{15}{20+5+15}\dot{\mathbf{I}}_2}{\dot{\mathbf{I}}_2} = 7.5\,\Omega$$

$$\begin{cases} \dot{U}_1 = Z_{11}\dot{I}_1 + Z_{12}\dot{I}_2 \\ \dot{U}_2 = Z_{21}\dot{I}_1 + Z_{22}\dot{I}_2 \end{cases}$$

$$\mathbf{Z_{22}} = \left.\frac{\dot{U}_2}{\dot{I}_2}\right|_{\dot{I}_1=0} = \frac{\dfrac{(20+5)\times 15}{20+5+15}\dot{I}_2}{\dot{I}_2} = 9.375\,\Omega$$

例2

图示电路，已知 $R=3\Omega,\omega L_1=\omega L_2=3\Omega,$ $\omega M=1\Omega$ ，求双口网络的**Z**参数。



解：直接列方程法。

在二个端口分别加电压源 $\dot{U}_1$ 和 $\dot{U}_2$ ，列回路电压方程

$$\begin{cases}\dot{U}_1=Z_{11}\dot{I}_1+Z_{12}\dot{I}_2\\ \dot{U}_2=Z_{21}\dot{I}_1+Z_{22}\dot{I}_2\end{cases}$$

$$\begin{cases}\dot{U}_1=j\omega L_1\dot{I}_1+j\omega M(\dot{I}_1+\dot{I}_2)+j\omega L_2(\dot{I}_1+\dot{I}_2)+j\omega M\dot{I}_1\\ \dot{U}_2=R\dot{I}_2+j\omega L_2(\dot{I}_1+\dot{I}_2)+j\omega M\dot{I}_1\end{cases}$$

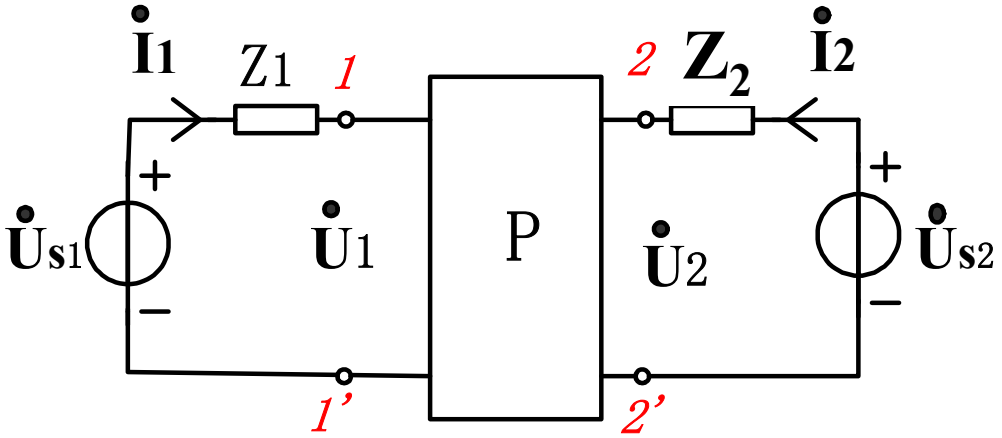
整理得 $$\begin{cases}\dot{U}_1=j\omega(L_1+L_2+2M)\dot{I}_1+j\omega(L_1+M)\dot{I}_2=j8\dot{I}_1+j4\dot{I}_2\\ \dot{U}_2=j\omega(L_2+M)\dot{I}_1+(j\omega L_2+R)\dot{I}_2=j4\dot{I}_1+(3+3j)\dot{I}_2\end{cases}$$

比较上式与**Z**参数方程，得

$$Z_{11}=j8\quad Z_{12}=j4\quad Z_{21}=j4\quad Z_{22}=3+3j$$

例3 包含双口网络的电路计算(应用回路电流法解题)。

如图电路，电源和阻抗参数及双口网络参数[Z]已知，求 $\dot{I}_1$ 和 $\dot{I}_2$ 。



解：包含双口网络的电路计算应用回路电流法。对网孔1和2列回路方程，得

$$\dot{U}_{s1} = Z_1\dot{I}_1 + \dot{U}_1$$
$$\dot{U}_{s2} = Z_2\dot{I}_2 + \dot{U}_2$$

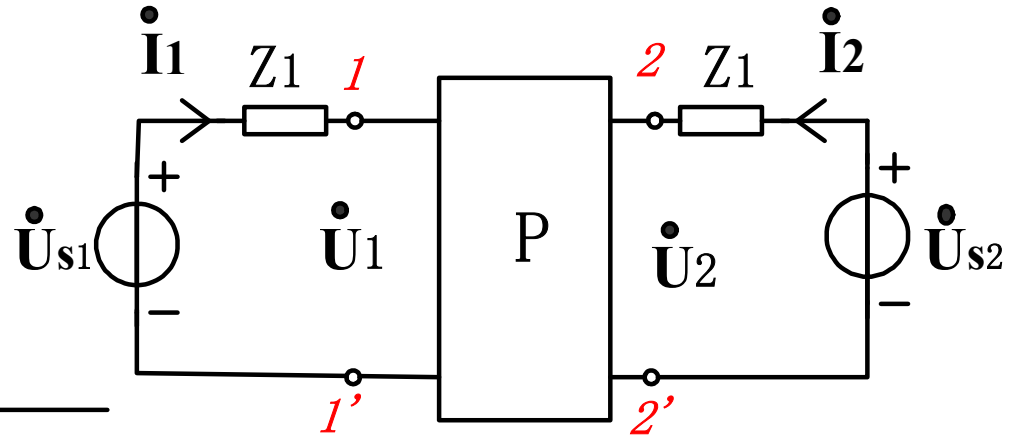
考虑到

$$\begin{cases} \dot{U}_1 = Z_{11}\dot{I}_1 + Z_{12}\dot{I}_2 \\ \dot{U}_2 = Z_{21}\dot{I}_1 + Z_{22}\dot{I}_2 \end{cases}$$

得

$$\dot{U}_{s1} = Z_1\dot{I}_1 + Z_{11}\dot{I}_1 + Z_{12}\dot{I}_2$$
$$\dot{U}_{s2} = Z_2\dot{I}_2 + Z_{21}\dot{I}_1 + Z_{22}\dot{I}_2$$

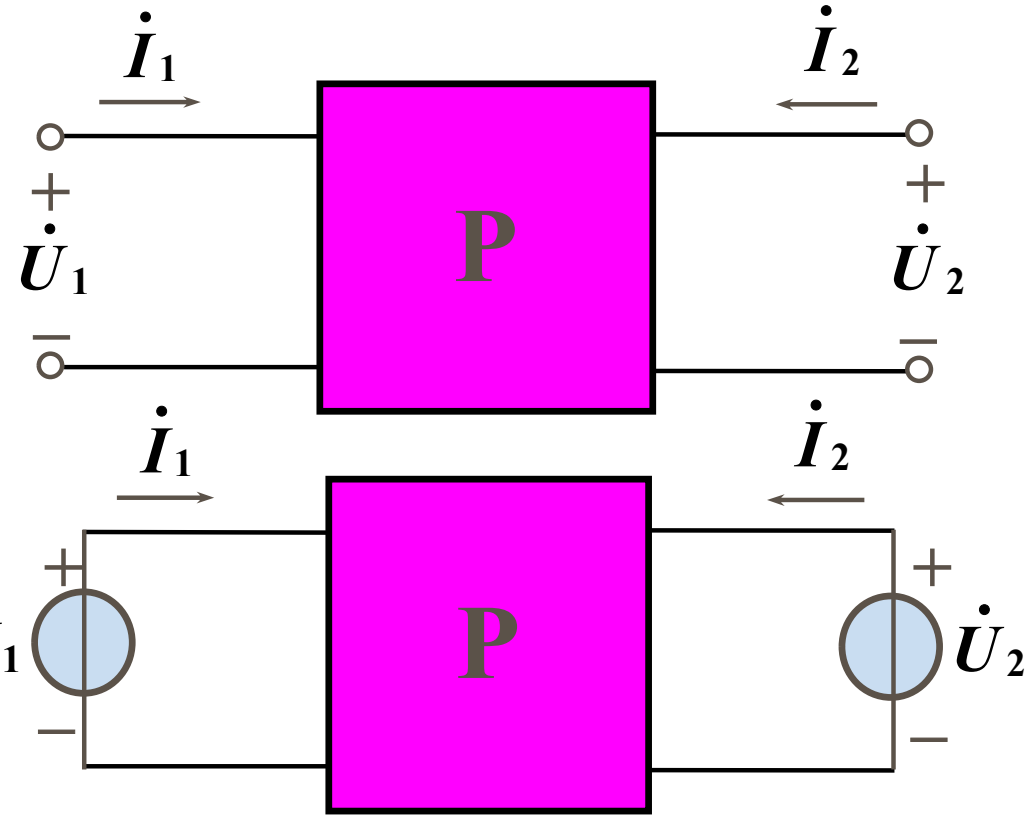
解得



$$\dot{I}_1 = \frac{(Z_2 + Z_{22})\dot{U}_{s1} - Z_{12}\dot{U}_{s2}}{(Z_2 + Z_{22})(Z_1 + Z_{11}) - Z_{12}Z_{21}}$$

$$\dot{I}_2 = \frac{(Z_1 + Z_{11})\dot{U}_{s2} - Z_{21}\dot{U}_{s1}}{(Z_1 + Z_{11})(Z_2 + Z_{22}) - Z_{12}Z_{21}}$$

10.3 双口网络的短路参数Y



1) 短路参数方程(用端口电压来表示端口电流)

$$\begin{cases} \dot{I}_1 = Y_{11}\dot{U}_1 + Y_{12}\dot{U}_2 \\ \dot{I}_2 = Y_{21}\dot{U}_1 + Y_{22}\dot{U}_2 \end{cases}$$

写成矩阵形式为:

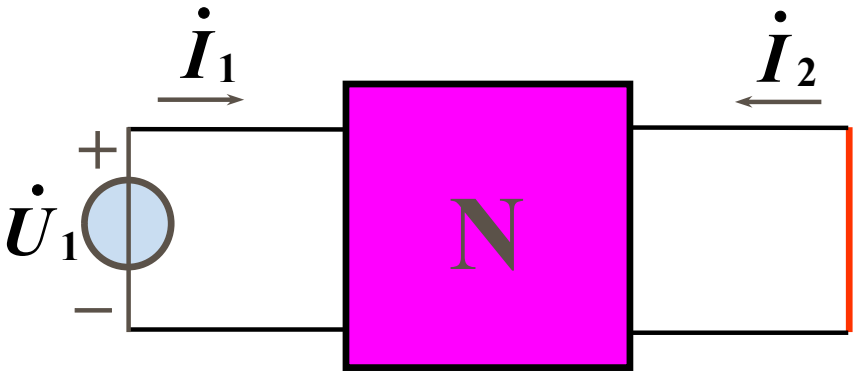
$$\begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix} \begin{bmatrix} \dot{U}_1 \\ \dot{U}_2 \end{bmatrix}$$

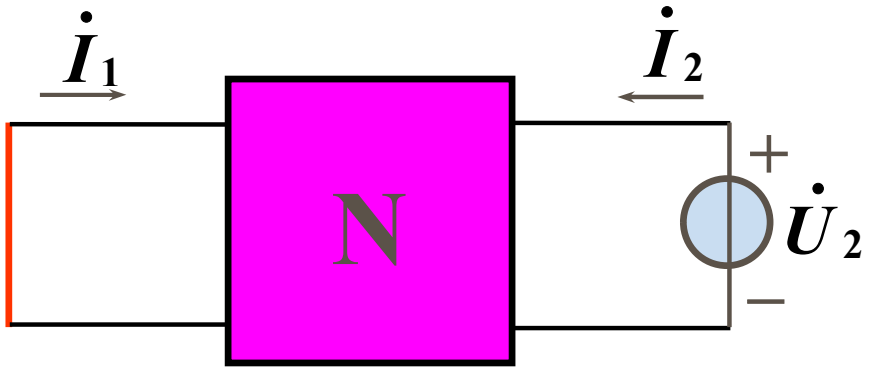
$$[Y] = \begin{bmatrix} Y_{11} & Y_{12} \\ Y_{21} & Y_{22} \end{bmatrix}$$

[Y] 称为Y 参数矩阵.

（2） Y参数的计算和测定

$$\begin{cases} \dot{I}_1 = Y_{11}\dot{U}_1 + Y_{12}\dot{U}_2 \\ \dot{I}_2 = Y_{21}\dot{U}_1 + Y_{22}\dot{U}_2 \end{cases}$$





由Y参数方程可得：

$$Y_{11} = \left.\frac{\dot{I}_1}{\dot{U}_1}\right|_{\dot{U}_2=0}$$

$$Y_{21} = \left.\frac{\dot{I}_2}{\dot{U}_1}\right|_{\dot{U}_2=0}$$

由Y参数方程可得：

$$Y_{12} = \left.\frac{\dot{I}_1}{\dot{U}_2}\right|_{\dot{U}_1=0}$$

$$Y_{22} = \left.\frac{\dot{I}_2}{\dot{U}_2}\right|_{\dot{U}_1=0}$$

Y参数都是短路情况下的导纳，又称为短路参数。

（3）互易双口　　$Y_{12} = Y_{21}$

对称双口　$Y_{12} = Y_{21}$，　$Y_{11} = Y_{22}$

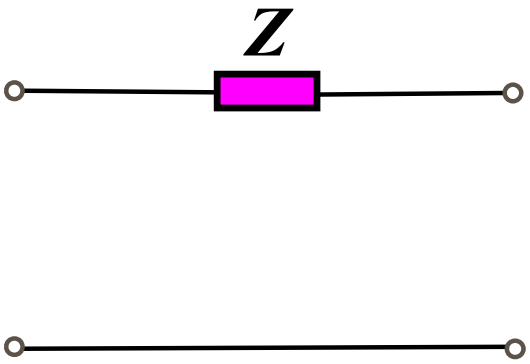
（4）　Z 参数矩阵与 Y 参数矩阵互为逆矩阵。

$$\begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix} = Y \begin{bmatrix} \dot{U}_1 \\ \dot{U}_2 \end{bmatrix}$$

$$\begin{bmatrix} \dot{U}_1 \\ \dot{U}_2 \end{bmatrix} = Z \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix}$$

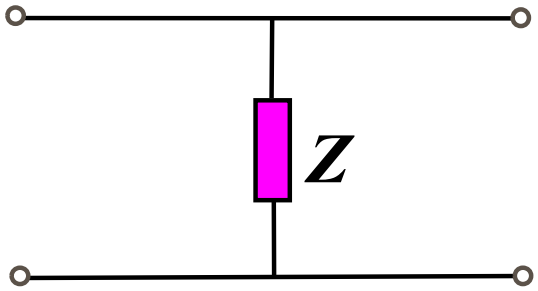
即：$\boldsymbol{Y} = \boldsymbol{Z}^{-1}$　　$\boldsymbol{Z} = \boldsymbol{Y}^{-1}$

（5）　并非所有的二端口均有 Z, Y 参数。



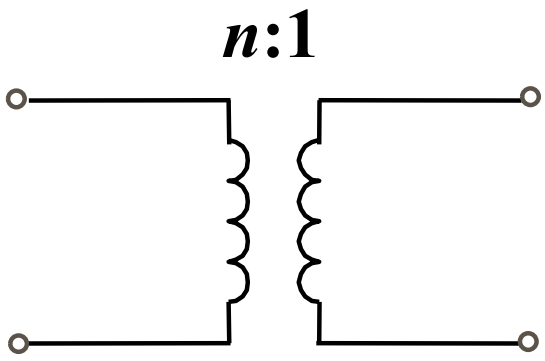
$$[Y]=\begin{bmatrix}\frac{1}{Z} & -\frac{1}{Z}\\ -\frac{1}{Z} & \frac{1}{Z}\end{bmatrix}$$

Z 不存在



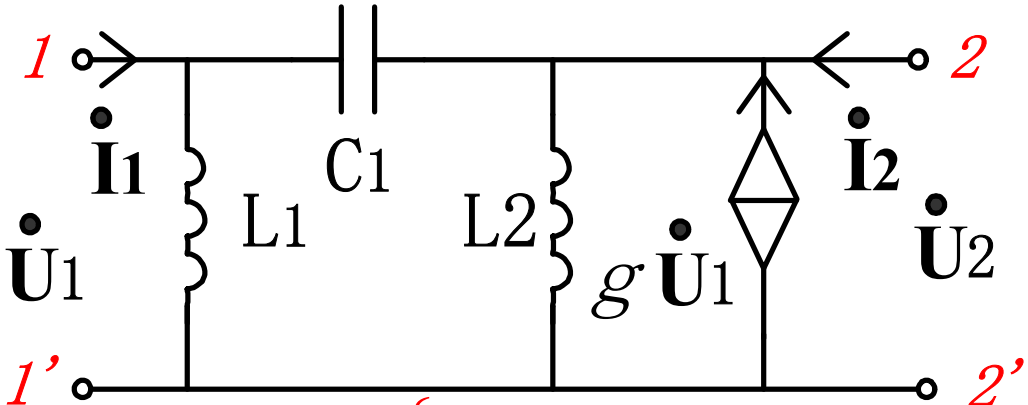
$$[Z]=\begin{bmatrix}Z & Z\\ Z & Z\end{bmatrix}$$

Y 不存在



Z,Y 均不存在

例1　设角频率为ω，求图示电路双口网络的短路参数。



解：直接列方程法。

加电压$\dot{U}_1$ $\dot{U}_2$，求 $\dot{I}_1$ $\dot{I}_2$

$$\dot{I}_1 = \frac{\dot{U}_1}{j\omega L_1} + j\omega C_1(\dot{U}_1 - \dot{U}_2)$$

$$\dot{I}_2 + g\dot{U}_1 + j\omega C_1(\dot{U}_1 - \dot{U}_2) = \frac{\dot{U}_2}{j\omega L_2}$$

整理得：$\dot{I}_1 = (\frac{1}{j\omega L_1} + j\omega C_1)\dot{U}_1 - j\omega C_1\dot{U}_2$

$$\dot{I}_2 = -(g + j\omega C_1)\dot{U}_1 + (j\omega C_1 + \frac{1}{j\omega L_2})\dot{U}_2$$

$$\begin{cases} \dot{I}_1 = Y_{11}\dot{U}_1 + Y_{12}\dot{U}_2 \\ \dot{I}_2 = Y_{21}\dot{U}_1 + Y_{22}\dot{U}_2 \end{cases}$$

$$Y_{11} = \frac{1}{j\omega L_1} + j\omega C_1$$

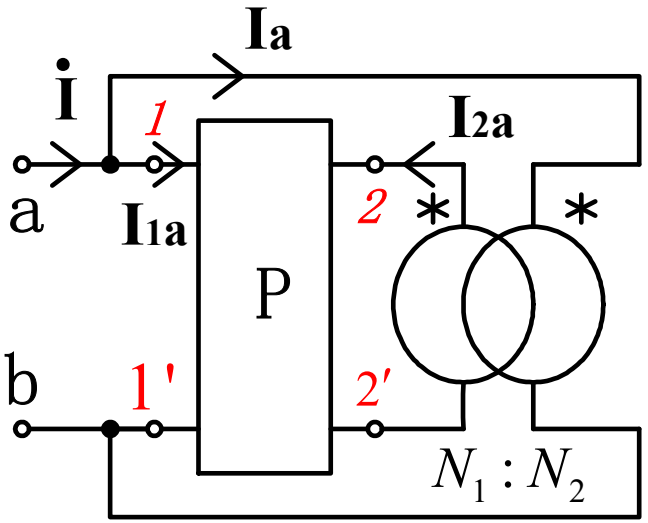
$$Y_{12} = -j\omega C_1$$

$$Y_{21} = -(g + j\omega C_1)$$

$$Y_{22} = j\omega C_1 + \frac{1}{j\omega L_2}$$

例2 如图电路，**P**为互易对称双口网络，两种联结时测得入端导纳分别为 Y_{ab} 和 Y_{cd}，理想变压器变比 $\dfrac{N_1}{N_2}=n$

求双口网络的**Y**参数。



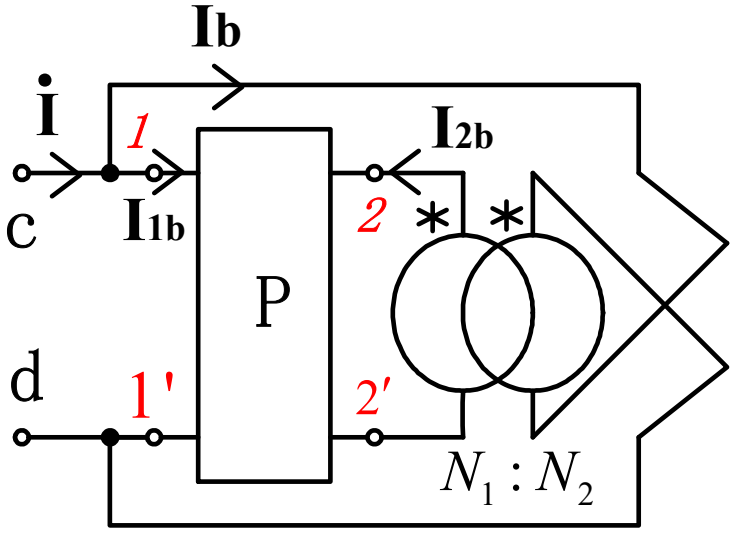
联结1

解：

$$\begin{cases}\dot{I}_1=Y_{11}\dot{U}_1+Y_{12}\dot{U}_2\\ \dot{I}_2=Y_{21}\dot{U}_1+Y_{22}\dot{U}_2\end{cases}$$

对于上图，

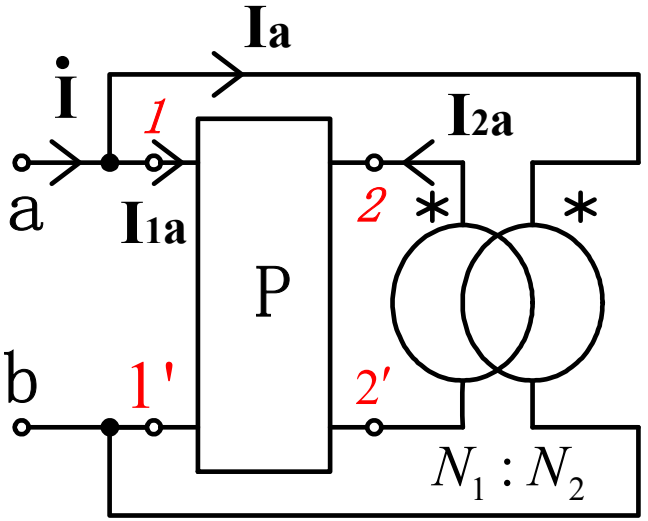
$$\begin{cases}\dot{I}_{1a}=Y_{11}\dot{U}_1+Y_{12}\dot{U}_2\\ \dot{I}_{2a}=Y_{21}\dot{U}_1+Y_{22}\dot{U}_2\end{cases}$$



联结2

由理想变压器关系，得：

$$\frac{\dot{U}_{22'}}{\dot{U}_{11'}}=\frac{N_1}{N_2}=n \qquad \frac{\dot{I}_{2a}}{\dot{I}_a}=\frac{1}{n}$$



由互易对称双口网络：

$$\begin{cases} Y_{11}=Y_{22}=Y_1 \\ Y_{12}=Y_{21}=Y_2 \end{cases}$$

$$\begin{cases} \dot{I}_{1a}=Y_1\dot{U}_1+Y_2\dot{U}_2 \\ \dot{I}_{2a}=Y_2\dot{U}_1+Y_1\dot{U}_2 \end{cases}$$

$$\dot{U}_{22'}=\dot{U}_2,$$

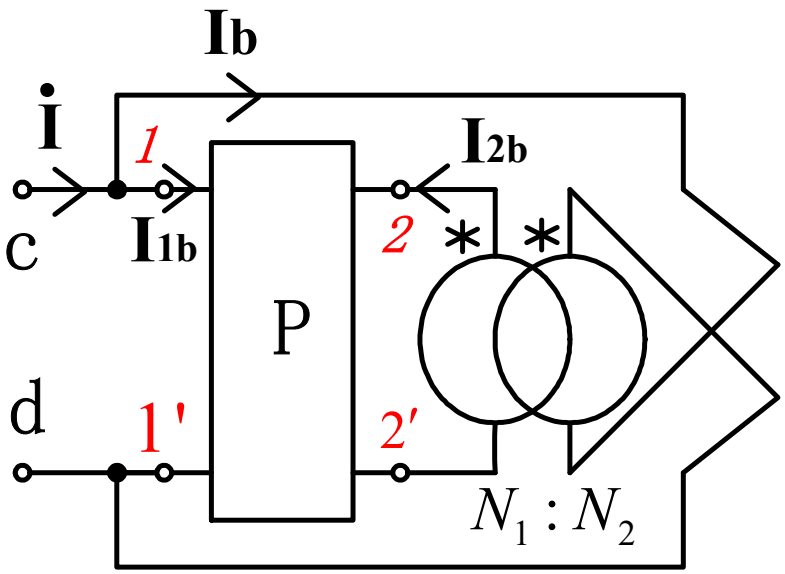
$$\dot{U}_{11'}=\dot{U}_1,$$

$$Y_{ab}=\frac{\dot{I}}{\dot{U}_{ab}}=\frac{\dot{I}_{1a}+\dot{I}_a}{\dot{U}_1}=\frac{Y_1\dot{U}_1+Y_2\dot{U}_2+n(Y_2\dot{U}_1+Y_1\dot{U}_2)}{\dot{U}_1}$$

$$=(1+n^2)Y_1+2nY_2$$

同理, 对于第二种连接, 有

$$\dot{U}_2 = -\frac{N_1}{N_2}\dot{U}_1 = -n\dot{U}_1 \qquad \dot{I}_b = -n\dot{I}_{2b}$$



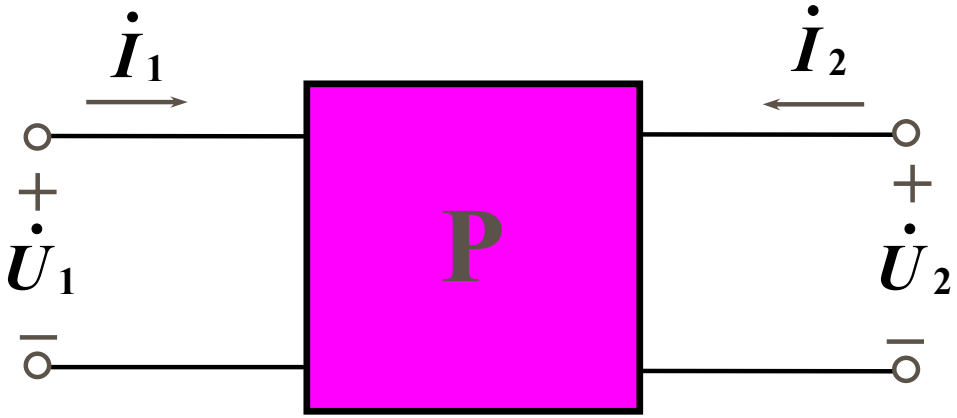
$$\begin{cases} \dot{\mathrm{I}}_{1b} = \mathrm{Y}_1\,\dot{\mathrm{U}}_1 + \mathrm{Y}_2\,\dot{\mathrm{U}}_2 \\ \dot{\mathrm{I}}_{2b} = \mathrm{Y}_2\,\dot{\mathrm{U}}_1 + \mathrm{Y}_1\,\dot{\mathrm{U}}_2 \end{cases}$$

$$Y_{cd} = \frac{\dot{I}}{\dot{U}_{cd}} = \frac{\dot{I}_{1b} + \dot{I}_b}{\dot{U}_1} = \frac{\mathrm{Y}_1\,\dot{\mathrm{U}}_1 + \mathrm{Y}_2\,\dot{\mathrm{U}}_2 - n(\mathrm{Y}_2\,\dot{\mathrm{U}}_1 + \mathrm{Y}_1\,\dot{\mathrm{U}}_2)}{\dot{U}_1}$$

$$= (1+n^2)Y_1 - 2nY_2$$

$$Y_1 = \frac{Y_{ab} + Y_{cd}}{2(1+n^2)} \qquad Y_2 = \frac{Y_{ab} - Y_{cd}}{4n}$$

10.4　双口网络的传输参数*T*



（1）*T* 参数

T 参数也称为传输参数

*Y*参数方程

$$\begin{cases} \dot{I}_1 = Y_{11}\dot{U}_1 + Y_{12}\dot{U}_2 & (1) \\ \dot{I}_2 = Y_{21}\dot{U}_1 + Y_{22}\dot{U}_2 & (2) \end{cases}$$

由(2)得：

$$\dot{U}_1 = -\frac{Y_{22}}{Y_{21}}\dot{U}_2 + \frac{1}{Y_{21}}\dot{I}_2 \quad (3)$$

将(3)代入(1)得：

$$\dot{I}_1 = \left(Y_{12} - \frac{Y_{11}Y_{22}}{Y_{21}}\right)\dot{U}_2 + \frac{Y_{11}}{Y_{21}}\dot{I}_2$$

即：$\begin{cases} \dot{U}_1 = A\dot{U}_2 + B(-\dot{I}_2) \\ \dot{I}_1 = C\dot{U}_2 + D(-\dot{I}_2) \end{cases}$

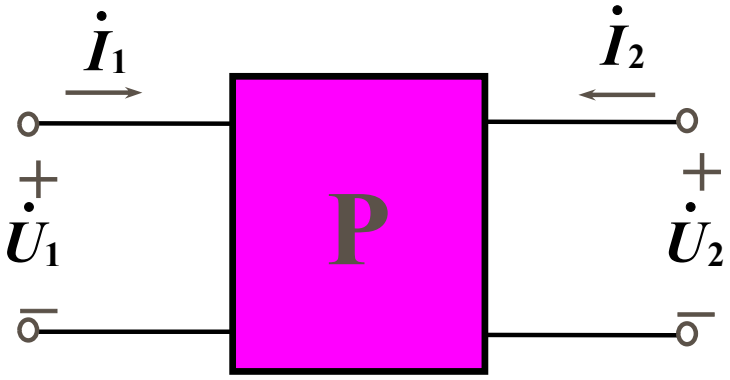
$$\begin{cases} \dot{U}_1 = -\dfrac{Y_{22}}{Y_{21}}\dot{U}_2 + \dfrac{1}{Y_{21}}\dot{I}_2 \\ \dot{I}_1 = \left(Y_{12} - \dfrac{Y_{11}Y_{22}}{Y_{21}}\right)\dot{U}_2 + \dfrac{Y_{11}}{Y_{21}}\dot{I}_2 \end{cases}$$

其中 $A = -\dfrac{Y_{22}}{Y_{21}}$ $B = \dfrac{-1}{Y_{21}}$

$C = \dfrac{Y_{12}Y_{21} - Y_{11}Y_{22}}{Y_{21}}$ $D = -\dfrac{Y_{11}}{Y_{21}}$

上述方程称为传输参数(*T* 参数)方程，其矩阵形式：

$$\begin{bmatrix} \dot{U}_1 \\ \dot{I}_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} \dot{U}_2 \\ -\dot{I}_2 \end{bmatrix}$$



(注意符号）

$$[T] = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$$

称为*T* 参数矩阵

（2） *T* 参数的计算或测定

$$\begin{cases}\dot{\mathbf{U}}_1 = \mathbf{A}\dot{\mathbf{U}}_2 + \mathbf{B}(-\dot{\mathbf{I}}_2) \\ \dot{\mathbf{I}}_1 = \mathbf{C}\dot{\mathbf{U}}_2 + \mathbf{D}(-\dot{\mathbf{I}}_2)\end{cases}$$

$$A = \left.\frac{\dot{U}_1}{\dot{U}_2}\right|_{\dot{I}_2=0} \qquad B = \left.\frac{\dot{U}_1}{-\dot{I}_2}\right|_{\dot{U}_2=0}$$

$$C = \left.\frac{\dot{I}_1}{\dot{U}_2}\right|_{\dot{I}_2=0} \qquad D = \left.\frac{\dot{I}_1}{-\dot{I}_2}\right|_{\dot{U}_2=0}$$

（3） 互易双口 $Y_{12} = Y_{21}$

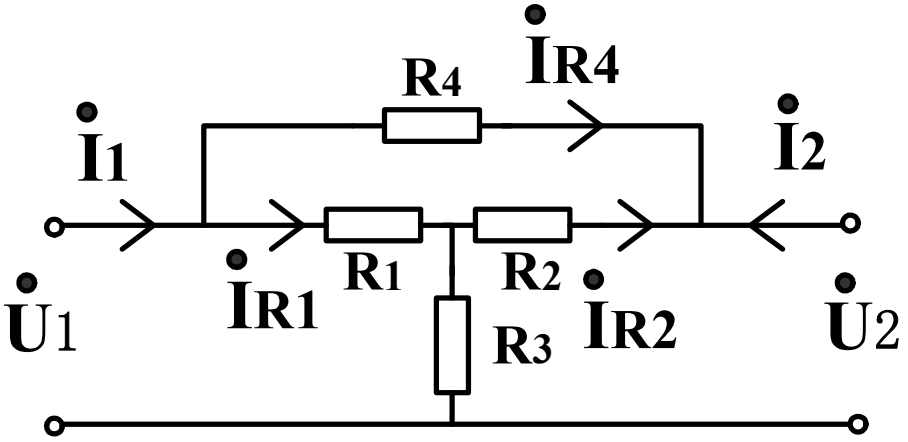
T 参数满足： $\mathbf{AD - BC = 1}$

对称双口 $Y_{11} = Y_{22}$ 则 $\mathbf{A = D}$

例**1**

$R_1 = 12\Omega, R_2 = 6\Omega, R_3 = 12\Omega, R_4 = 6\Omega,$

求传输参数。



解: 定义法或直接列方程法（略）

$$\begin{cases} \dot{U}_1 = A\dot{U}_2 + B(-\dot{I}_2) \\ \dot{I}_1 = C\dot{U}_2 + D(-\dot{I}_2) \end{cases}$$

$$A = \frac{6}{5} \qquad C = \frac{1}{15}$$

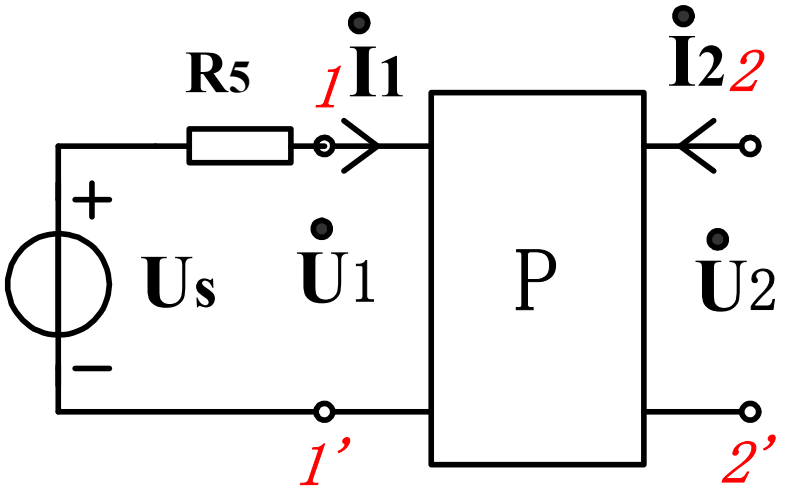
$$B = \frac{24}{5} \qquad D = \frac{1}{2}$$

例2

已知二端口网络传输参数，$T=\begin{bmatrix} \frac{3}{2} & 12 \\ \frac{1}{6} & 2 \end{bmatrix}$

$\dot{U}_s=6\text{V}, R_5=3\Omega$

试求2-2’端戴维南等效电路。



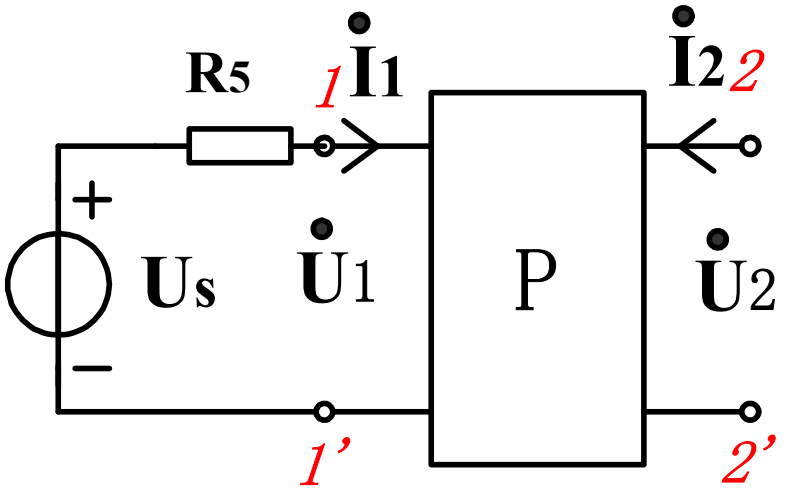
解：*T*参数方程为 $\begin{cases} \dot{U}_1=A\dot{U}_2+B(-\dot{I}_2) \\ \dot{I}_1=C\dot{U}_2+D(-\dot{I}_2) \end{cases}$

2-2’开路时，1-1’入端阻抗，

$$Z_{01}=\left.\frac{\dot{U}_1}{\dot{I}_1}\right|_{\dot{I}_2=0}=\frac{A}{C}=9\Omega \qquad U_{01}=Us\frac{Z_{01}}{R_5+Z_{01}}=4.5\text{V}$$

$$U_{02}=\frac{U_{01}}{A}=3\text{V} \quad \text{（开路电压）} \qquad U_0=U_{02}=3\text{V}$$

$$\begin{cases}\dot{U}_1 = A\dot{U}_2 + B(-\dot{I}_2) \\ \dot{I}_1 = C\dot{U}_2 + D(-\dot{I}_2)\end{cases}$$



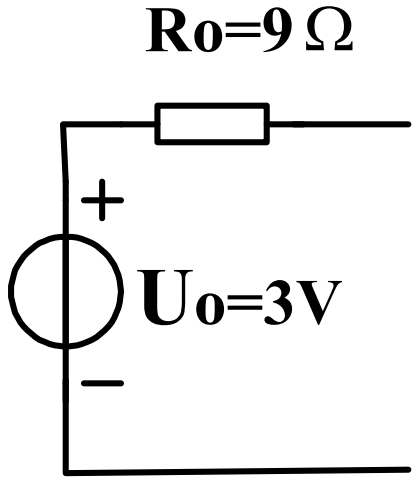
$2-2'$ 短路时，$1-1'$ 入端阻抗

$$Z_{s1} = \left.\frac{\dot{U}_1}{\dot{I}_1}\right|_{\dot{U}_2=0} = \frac{B}{D} = 6\Omega$$

$$U_{s1} = Us\frac{Z_{s1}}{R_5 + Z_{s1}} = 4\text{V} \qquad -I_{S2} = \frac{U_{S1}}{B} = \frac{1}{3}\text{A}$$

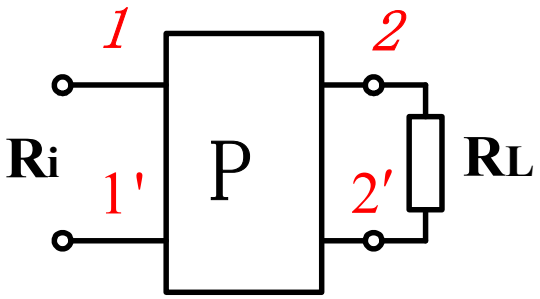
2-2’端入端电阻 $R_0 = \dfrac{U_0}{-I_{S2}} = 9\Omega$

等效电路：



例3 图示电路,**P**为线性无源电阻双口网络,已知输入电阻

$R_i = 10 - \dfrac{100}{R_L + 12}\Omega$, R_L 为任意电阻. 求**P**的传输参数**T.**



解: $R_L = \dfrac{\dot{U}_2}{-\dot{I}_2}$

$$R_i = \frac{\dot{U}_1}{\dot{I}_1} = \frac{A\dot{U}_2 + B(-\dot{I}_2)}{C\dot{U}_2 + D(-\dot{I}_2)} = \frac{AR_L + B}{CR_L + D}$$

$$\begin{cases} \dot{U}_1 = A\dot{U}_2 + B(-\dot{I}_2) \\ \dot{I}_1 = C\dot{U}_2 + D(-\dot{I}_2) \end{cases}$$

题中 $R_i = 10 - \dfrac{100}{R_L + 12} = \dfrac{10R_L + 20}{R_L + 12}$

$A = 10k \quad C = k$

$B = 20k \quad D = 12k$

$AD - BC = 1 \longrightarrow 100k^2 = 1$

$k = \pm 0.1$ 取 $k = 0.1$

得: $A = 1, B = 2, C = 0.1, D = 1.2$

10.5　双口网络的混合参数H

H 参数也称为混合参数，常用于晶体管等效电路。

（1） *H* 参数

$$\begin{cases} \dot{U}_1 = H_{11}\dot{I}_1 + H_{12}\dot{U}_2 \\ \dot{I}_2 = H_{21}\dot{I}_1 + H_{22}\dot{U}_2 \end{cases}$$

矩阵形式:

$$\begin{bmatrix} \dot{U}_1 \\ \dot{I}_2 \end{bmatrix} = \begin{bmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{bmatrix} \begin{bmatrix} \dot{I}_1 \\ \dot{U}_2 \end{bmatrix}$$

（2） *H* 参数的计算与测定

$$\begin{cases}\dot{U}_1 = H_{11}\dot{I}_1 + H_{12}\dot{U}_2 \\ \dot{I}_2 = H_{21}\dot{I}_1 + H_{22}\dot{U}_2\end{cases}$$

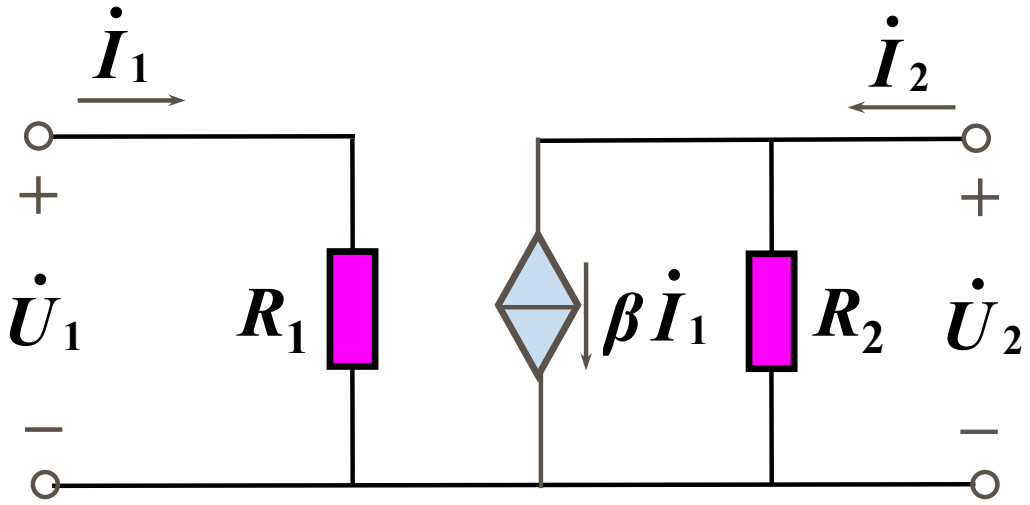
$$H_{11} = \left.\frac{\dot{U}_1}{\dot{I}_1}\right|_{\dot{U}_2=0} \qquad H_{12} = \left.\frac{\dot{U}_1}{\dot{U}_2}\right|_{\dot{I}_1=0}$$

$$H_{21} = \left.\frac{\dot{I}_2}{\dot{I}_1}\right|_{\dot{U}_2=0} \qquad H_{22} = \left.\frac{\dot{I}_2}{\dot{U}_2}\right|_{\dot{I}_1=0}$$

（3） 互易双口 $H_{12} = -H_{21}$

对称双口 $H_{11}H_{22} - H_{12}H_{21} = 1$

例: 图示电路为晶体管的小信号工作简化电路, 求H参数的值。

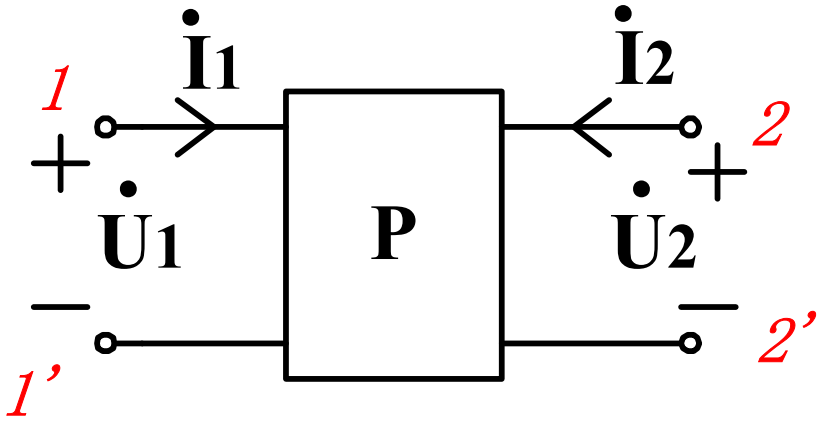


$$\begin{cases} \dot{U}_1 = H_{11}\dot{I}_1 + H_{12}\dot{U}_2 \\ \dot{I}_2 = H_{21}\dot{I}_1 + H_{22}\dot{U}_2 \end{cases}$$

$H_{11} = \left.\dfrac{\dot{U}_1}{\dot{I}_1}\right|_{\dot{U}_2=0} = R_1$ 输入电阻　　$H_{12} = \left.\dfrac{\dot{U}_1}{\dot{U}_2}\right|_{\dot{I}_1=0} = 0$

$H_{21} = \left.\dfrac{\dot{I}_2}{\dot{I}_1}\right|_{\dot{U}_2=0} = \beta$ 电流放大倍数　　$H_{22} = \left.\dfrac{\dot{I}_2}{\dot{U}_2}\right|_{\dot{I}_1=0} = \dfrac{1}{R_2}$ 输出电导

(4) 混合参数与Y参数的互换计算



$$\begin{cases} \dot{I}_1 = Y_{11}\dot{U}_1 + Y_{12}\dot{U}_2 \\ \dot{I}_2 = Y_{21}\dot{U}_1 + Y_{22}\dot{U}_2 \end{cases}$$

$$\downarrow$$

$$\begin{cases} \dot{U}_1 = \dfrac{1}{Y_{11}}\dot{I}_1 - \dfrac{Y_{12}}{Y_{11}}\dot{U}_2 \\ \dot{I}_2 = Y_{21}\left(\dfrac{1}{Y_{11}}\dot{I}_1 - \dfrac{Y_{12}}{Y_{11}}\dot{U}_2\right) + Y_{22}\dot{U}_2 \end{cases}$$

$$\downarrow$$

$$\begin{cases} \dot{U}_1 = \dfrac{1}{Y_{11}}\dot{I}_1 - \dfrac{Y_{12}}{Y_{11}}\dot{U}_2 \\ \dot{I}_2 = Y_{21}\dfrac{1}{Y_{11}}\dot{I}_1 + \left(Y_{22} - \dfrac{Y_{12}}{Y_{11}}\right)\dot{U}_2 \end{cases}$$

$$\longrightarrow$$

$$H_{11} = \frac{1}{Y_{11}}, \qquad H_{12} = -\frac{Y_{12}}{Y_{11}}$$

$$H_{21} = Y_{21}\frac{1}{Y_{11}}, \quad H_{22} = \left(Y_{22} - \frac{Y_{12}}{Y_{11}}\right)$$

$$\begin{cases} \dot{U}_1 = H_{11}\dot{I}_1 + H_{12}\dot{U}_2 \\ \dot{I}_2 = H_{21}\dot{I}_1 + H_{22}\dot{U}_2 \end{cases}$$

双口网络参数小结

1）开路参数

$$Z=\begin{cases}\dot{U}_1=Z_{11}\dot{I}_1+Z_{12}\dot{I}_2\\ \dot{U}_2=Z_{21}\dot{I}_1+Z_{22}\dot{I}_2\end{cases}$$

2）短路参数

$$Y=\begin{cases}\dot{I}_1=Y_{11}\dot{U}_1+Y_{12}\dot{U}_2\\ \dot{I}_2=Y_{21}\dot{U}_1+Y_{22}\dot{U}_2\end{cases}$$

3）传输参数

$$T=\begin{cases}\dot{U}_1=A\dot{U}_2+B(-\dot{I}_2)\\ \dot{I}_1=C\dot{U}_2+D(-\dot{I}_2)\end{cases}$$

4）混合参数

$$\begin{cases}\dot{U}_1=H_{11}\dot{I}_1+H_{12}\dot{U}_2\\ \dot{I}_2=H_{21}\dot{I}_1+H_{22}\dot{U}_2\end{cases}$$

①参数计算式可直接由定义式得出。

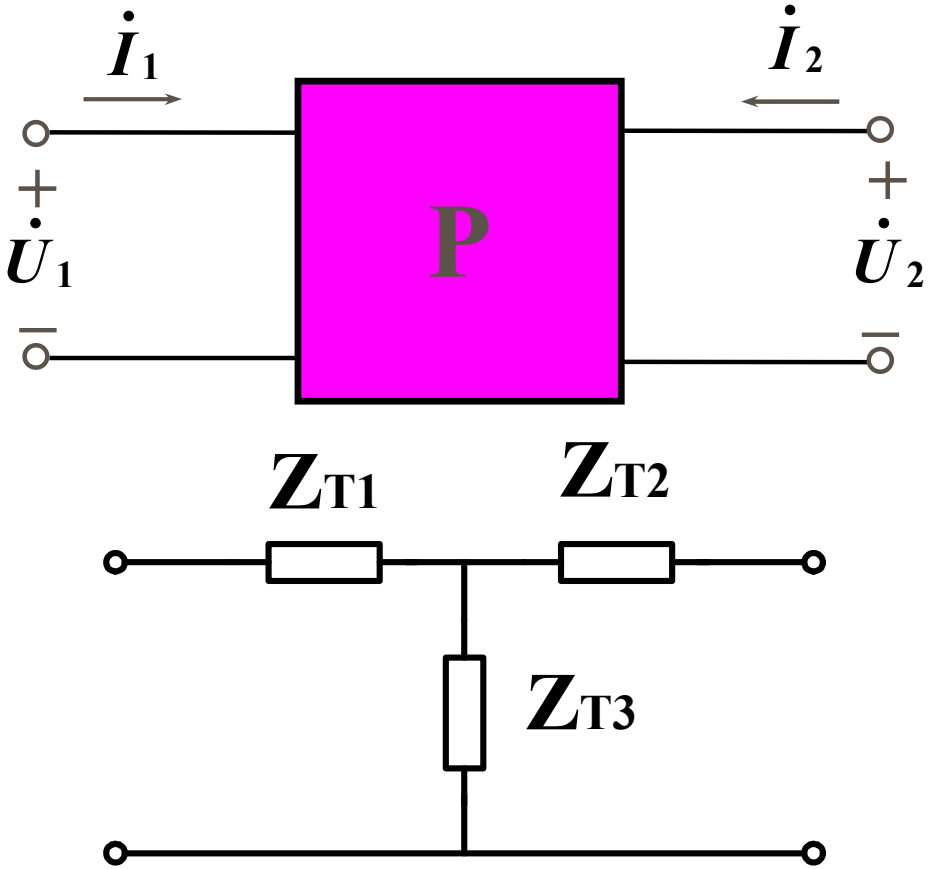
②参数互换式可由定义式解出相应变量后得出。

10.7 双口网络的等效

1. 互易双口网络的等效电路

1）T型等效电路

互易双口网络可用三个元件组成的“T”型等效电路来替代。



“T”型等效电路

a) 已知Z参数，计算“T”型电路参数：

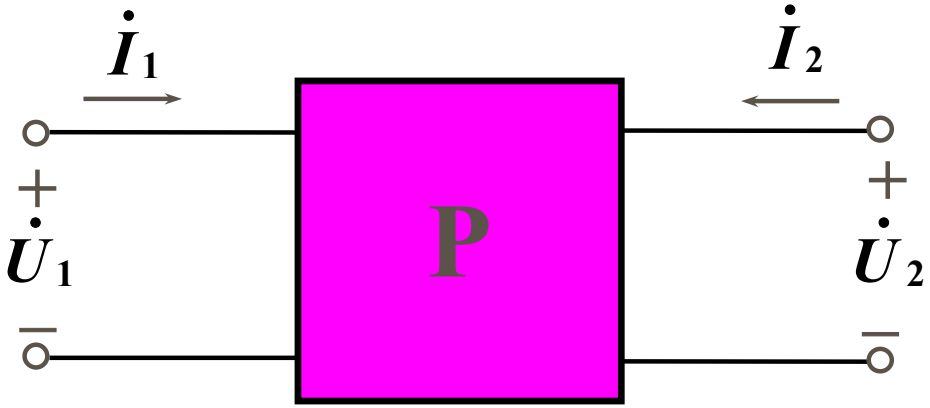
$$Z_{11}=Z_{T1}+Z_{T3},Z_{22}=Z_{T2}+Z_{T3},$$

$$Z_{12}=Z_{21}=Z_{T3},$$

得“T”型电路参数

$$\begin{cases} Z_{T1}=Z_{11}-Z_{12} \\ Z_{T2}=Z_{22}-Z_{12} \\ Z_{T3}=Z_{12} \end{cases}$$

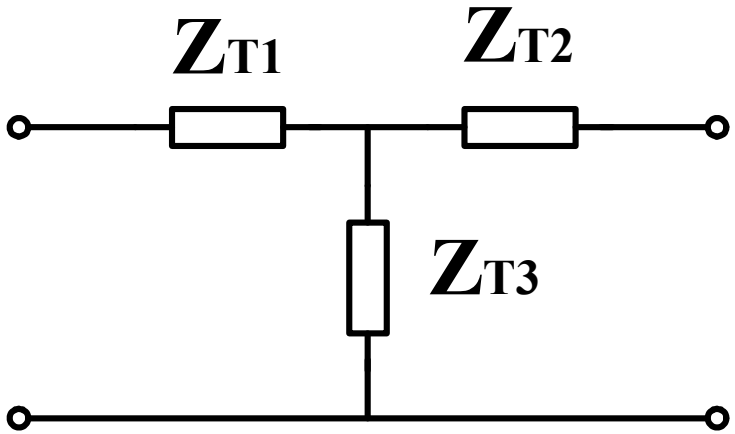
b)已知传输参数T，计算“T”型电路参数



$$A=\left.\frac{\dot{U}_1}{\dot{U}_2}\right|_{\dot{I}_2=0}=\frac{Z_{T1}+Z_{T3}}{Z_{T3}}$$

$$C=\left.\frac{\dot{I}_1}{\dot{U}_2}\right|_{\dot{I}_2=0}=\frac{1}{Z_{T3}}$$

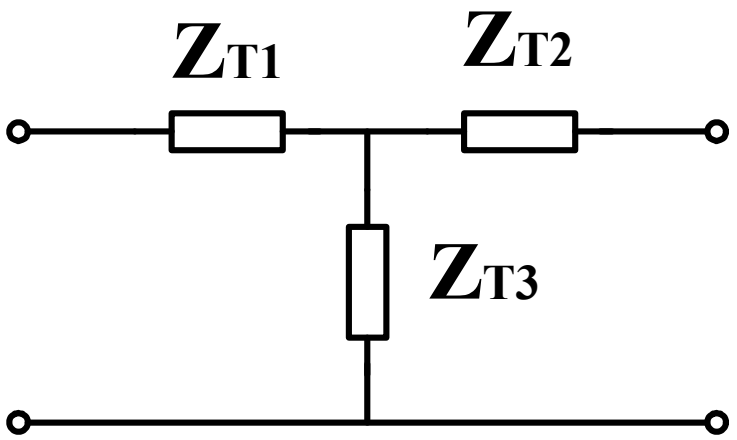
$$D=\left.\frac{\dot{I}_1}{-\dot{I}_2}\right|_{\dot{U}_2=0}=\frac{Z_{T2}+Z_{T3}}{Z_{T3}}$$



解得 $Z_{T1}=\frac{A-1}{C}$　　$Z_{T2}=\frac{D-1}{C}$　　$Z_{T3}=\frac{1}{C}$

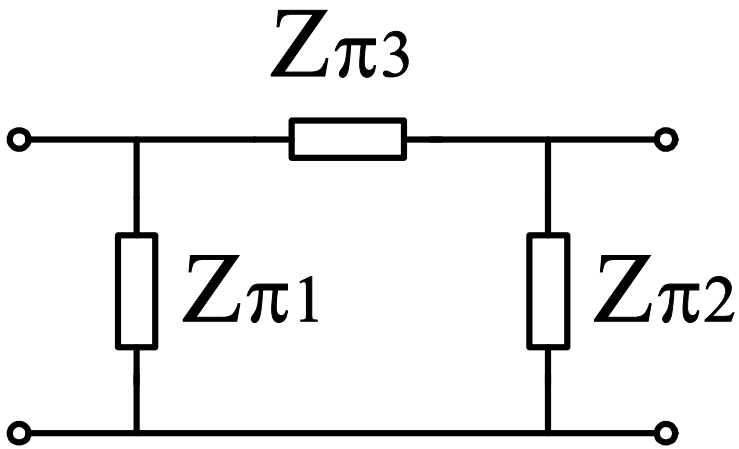
2）双口网络“π”型等效

互易双口网络可等效成三个元件的“π”型电路



等效参数计算（T→π相当于 Y→Δ）

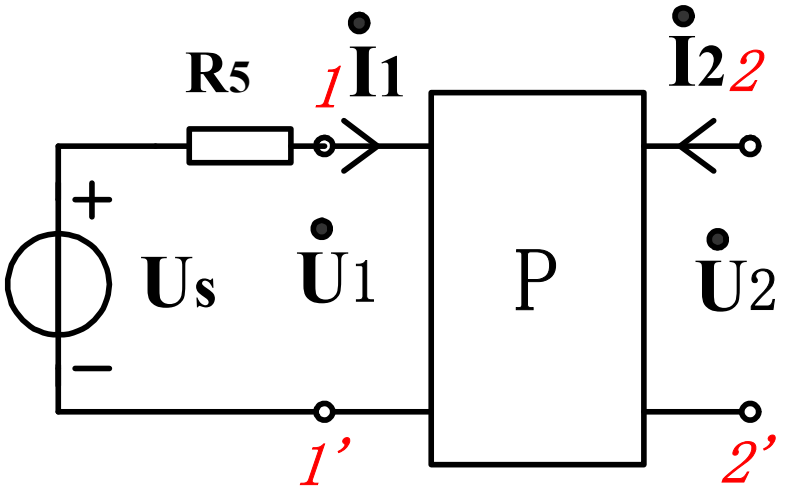
$$\begin{cases} Z_{\pi 1} = \dfrac{B}{D-1} \\ Z_{\pi 2} = \dfrac{B}{A-1} \\ Z_{\pi 3} = B \end{cases}$$



例

已知二端口网络传输参数，$T=\begin{bmatrix} \frac{3}{2} & 12 \\ \frac{1}{6} & 2 \end{bmatrix}$

$U_S=6\text{V}, R_5=3\Omega$

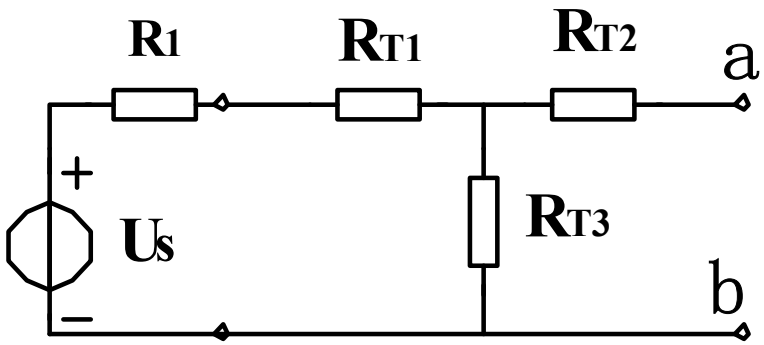


试求**2-2’**端戴维南等效电路。

解**:** 对**P**作“**T**”型等效替换

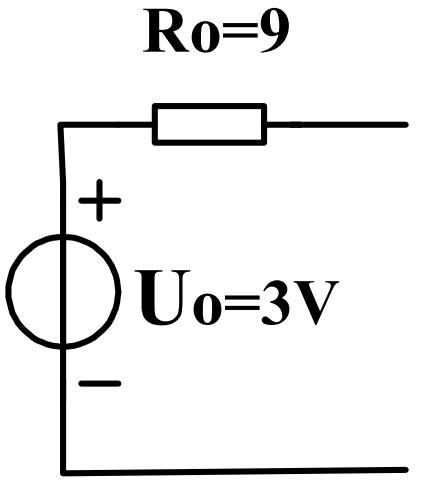
$$R_{T1}=\frac{A-1}{C}=3\Omega \qquad R_{T2}=\frac{D-1}{C}=6\Omega$$

$$R_{T3}=\frac{1}{C}=6\Omega$$



开路电压 $U_0=U_S\dfrac{R_{T3}}{R_1+R_{T1}+R_{T3}}=3\text{V}$

入端电阻 $R_0=R_{T2}+\dfrac{R_{T3}(R_1+R_{T1})}{R_1+R_{T1}+R_{T3}}=9\Omega$



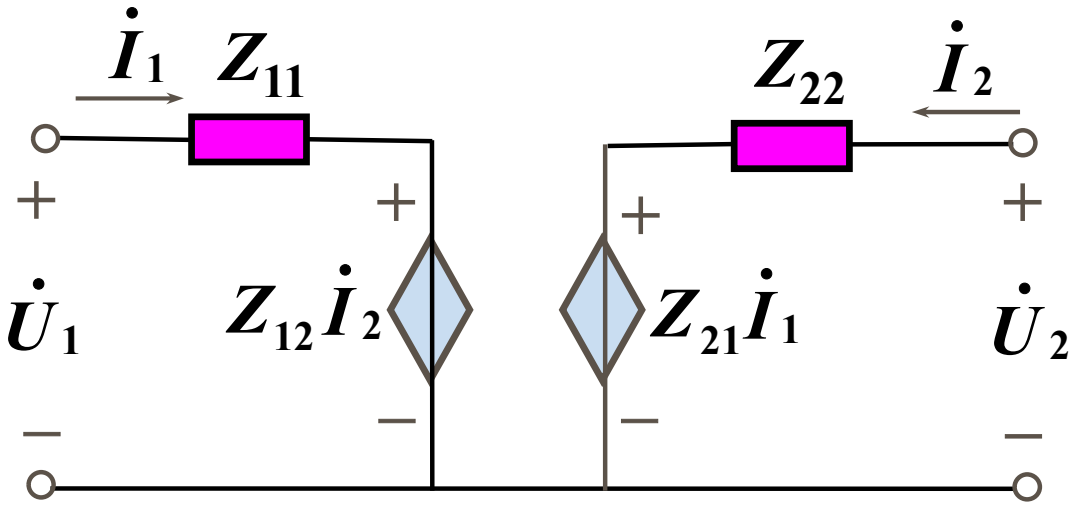
2. 一般二端口的等效电路(含受控源二端口)

方法：直接由参数方程得到等效电路。

例1.

$$\begin{cases} \dot{U}_1 = Z_{11}\dot{I}_1 + Z_{12}\dot{I}_2 \\ \dot{U}_2 = Z_{21}\dot{I}_1 + Z_{22}\dot{I}_2 \end{cases}$$

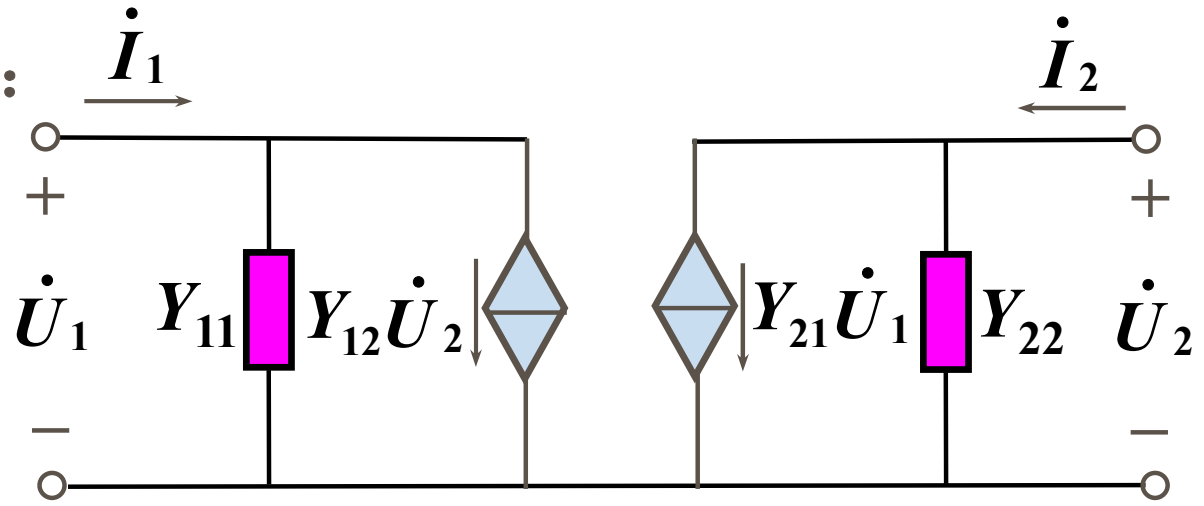
等效电路为：



若已知Y参数

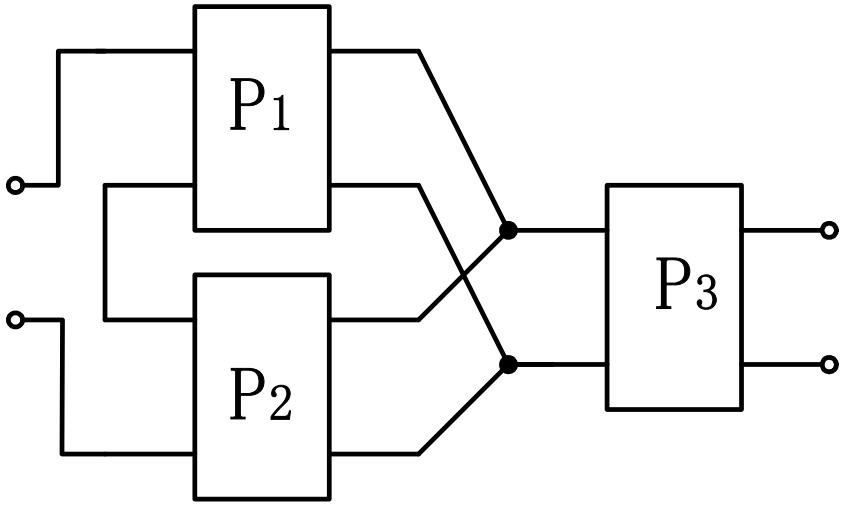
$$\begin{cases}\dot{I}_1 = Y_{11}\dot{U}_1 + Y_{12}\dot{U}_2 \\ \dot{I}_2 = Y_{21}\dot{U}_1 + Y_{22}\dot{U}_2\end{cases}$$

等效电路为：

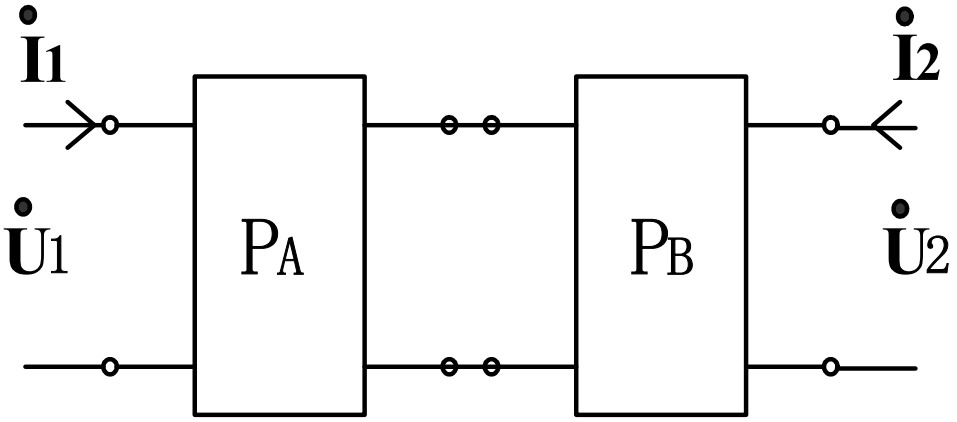


10.8 双口网络的连接

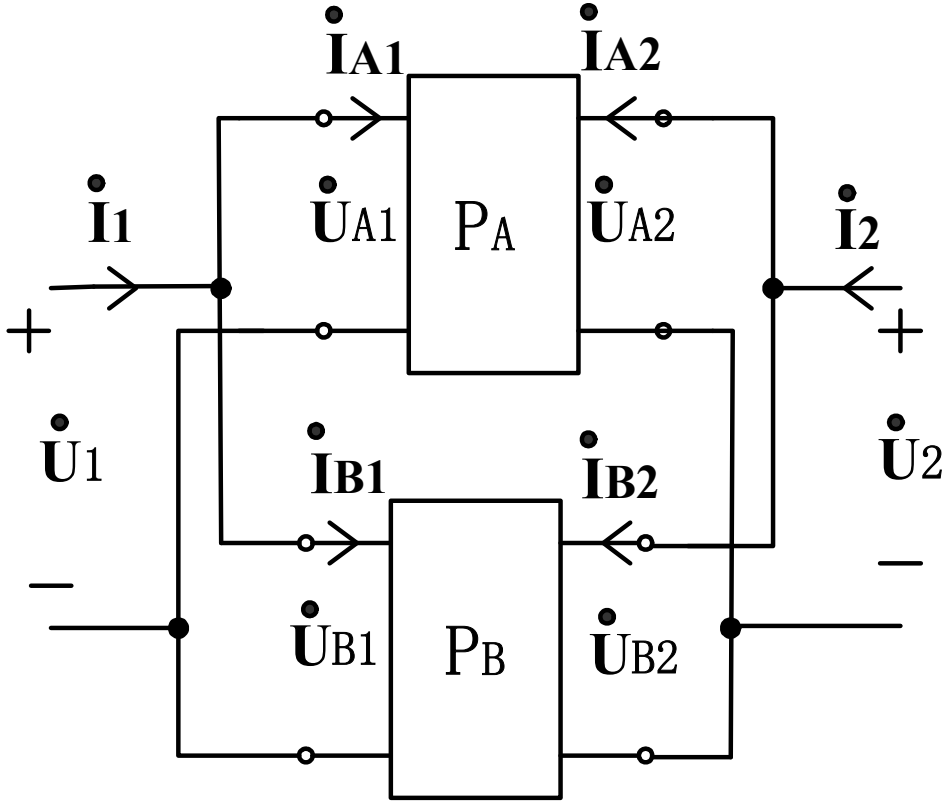
双口网络连接时的参数计算.



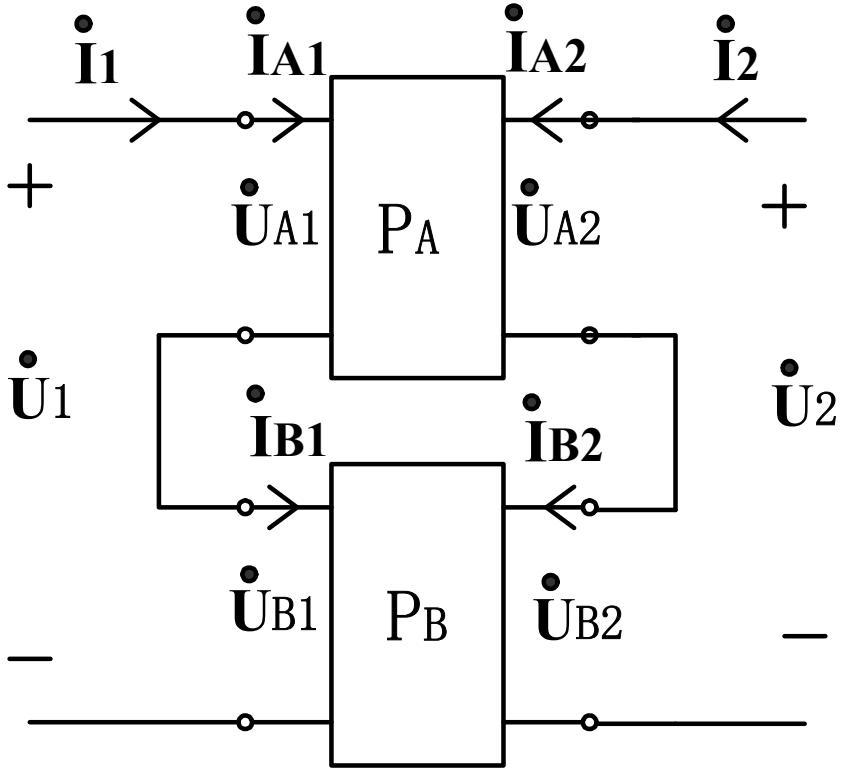
1. 双口网络的连接形式介绍



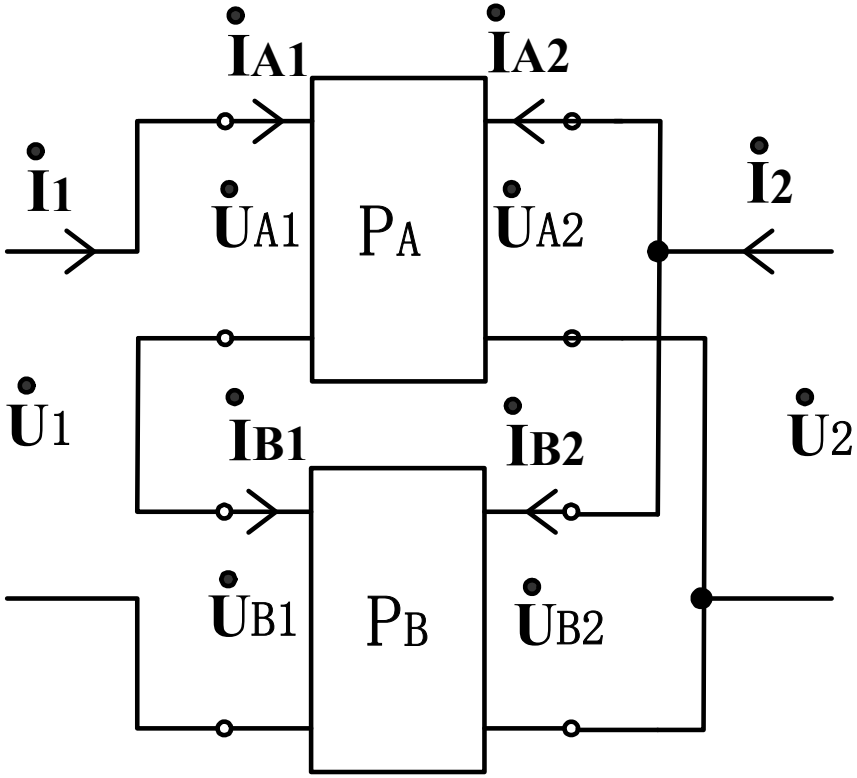
级联


İA1
İA2
İ1
U̇A1
PA
U̇A2
İ2
+
+
U̇1
İB1
İB2
U̇2
−
−
U̇B1
PB
U̇B2

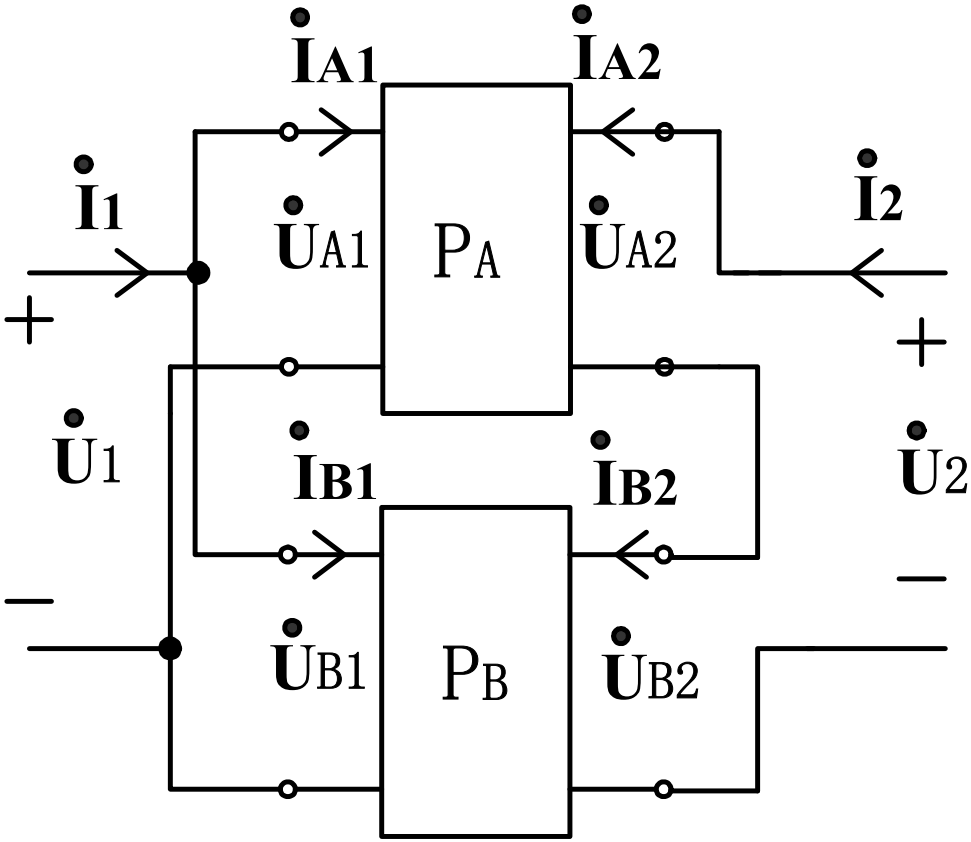
并联


İ1
İA1
İA2
İ2
+
U̇A1
PA
U̇A2
+
U̇1
İB1
İB2
U̇2
U̇B1
PB
U̇B2
−
−

串联

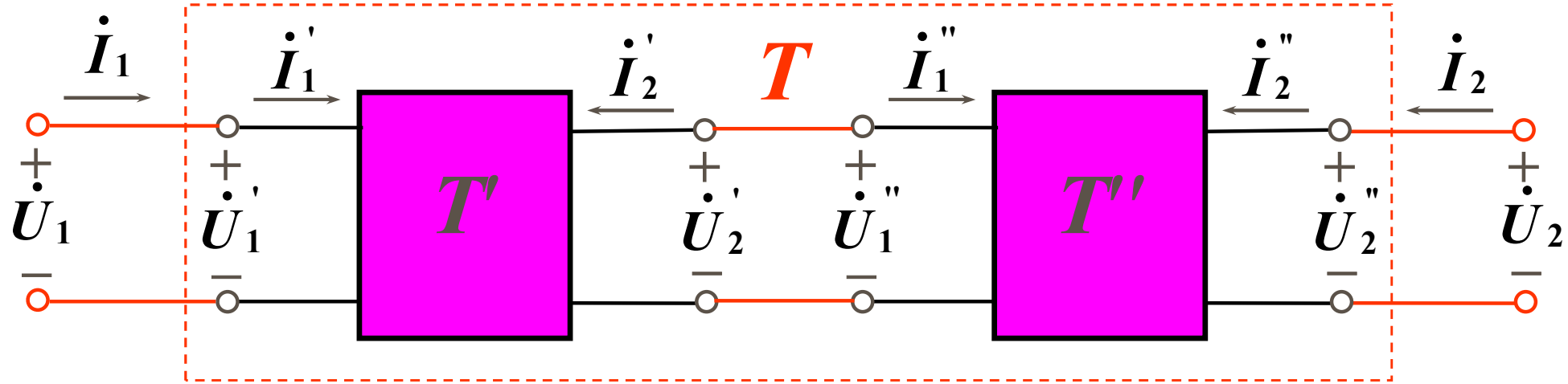

$\dot{I}_{A1}$
$\dot{I}_{A2}$
$\dot{I}_1$
$\dot{U}_{A1}$
P_A
$\dot{U}_{A2}$
$\dot{I}_2$
$\dot{U}_1$
$\dot{I}_{B1}$
$\dot{I}_{B2}$
$\dot{U}_2$
$\dot{U}_{B1}$
P_B
$\dot{U}_{B2}$

串并联


$\dot{I}_{A1}$
$\dot{I}_{A2}$
$\dot{I}_1$
$\dot{U}_{A1}$
P_A
$\dot{U}_{A2}$
$\dot{I}_2$
+
+
$\dot{U}_1$
$\dot{I}_{B1}$
$\dot{I}_{B2}$
$\dot{U}_2$
−
−
$\dot{U}_{B1}$
P_B
$\dot{U}_{B2}$

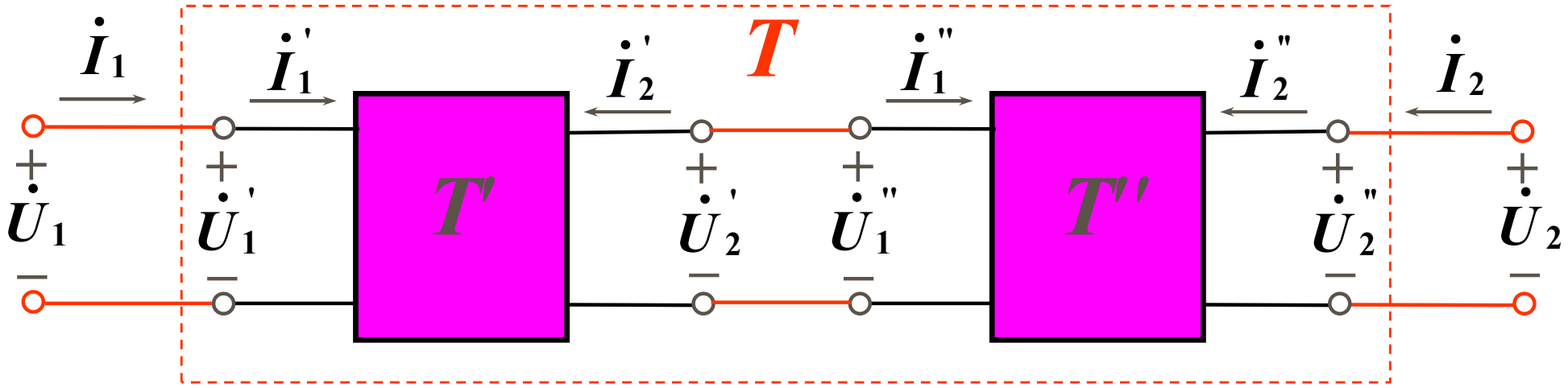
并串联

1. 级联(链联)：联接方式如下图



设 $[T']=\begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix}$ $[T'']=\begin{bmatrix} A'' & B'' \\ C'' & D'' \end{bmatrix}$

即 $\begin{bmatrix} \dot{U}_1' \\ \dot{I}_1' \end{bmatrix}=\begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix}\begin{bmatrix} \dot{U}_2' \\ -\dot{I}_2' \end{bmatrix}$ $\begin{bmatrix} \dot{U}_1'' \\ \dot{I}_1'' \end{bmatrix}=\begin{bmatrix} A'' & B'' \\ C'' & D'' \end{bmatrix}\begin{bmatrix} \dot{U}_2'' \\ -\dot{I}_2'' \end{bmatrix}$

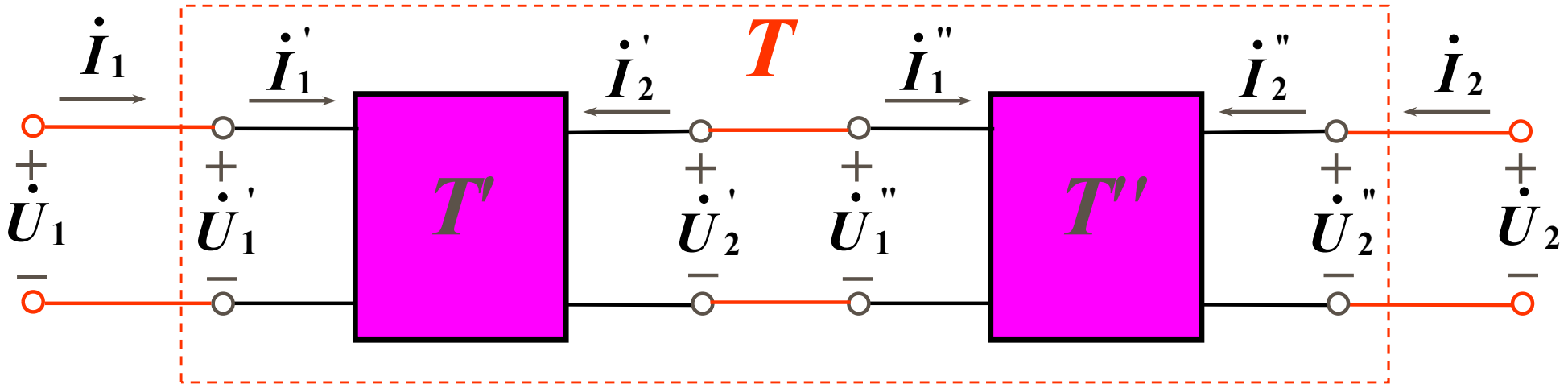


级联后

$$\begin{bmatrix}\dot{U}_1\\ \dot{I}_1\end{bmatrix}=\begin{bmatrix}\dot{U}_1'\\ \dot{I}_1'\end{bmatrix}=\begin{bmatrix}A' & B'\\ C' & D'\end{bmatrix}\begin{bmatrix}\dot{U}_2'\\ -\dot{I}_2'\end{bmatrix}$$

$$=\begin{bmatrix}A' & B'\\ C' & D'\end{bmatrix}\begin{bmatrix}A'' & B''\\ C'' & D''\end{bmatrix}\begin{bmatrix}\dot{U}_2\\ -\dot{I}_2\end{bmatrix}=\begin{bmatrix}A & B\\ C & D\end{bmatrix}\begin{bmatrix}\dot{U}_2\\ -\dot{I}_2\end{bmatrix}$$

T’ T” T



则

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} \begin{bmatrix} A'' & B'' \\ C'' & D'' \end{bmatrix} = \begin{bmatrix} A'A'' + B'C'' & A'B'' + B'D'' \\ C'A'' + D'C'' & C'B'' + D'D'' \end{bmatrix}$$

即：$$[T] = [T'][T'']$$

结论：级联后所得复合双口网络T 参数矩阵等于级联的双口网络T 参数矩阵相乘。上述结论可推广到n个双口网络级联的关系。

注意：

(1) 级联时T参数是矩阵相乘的关系，不是对应元素相乘。

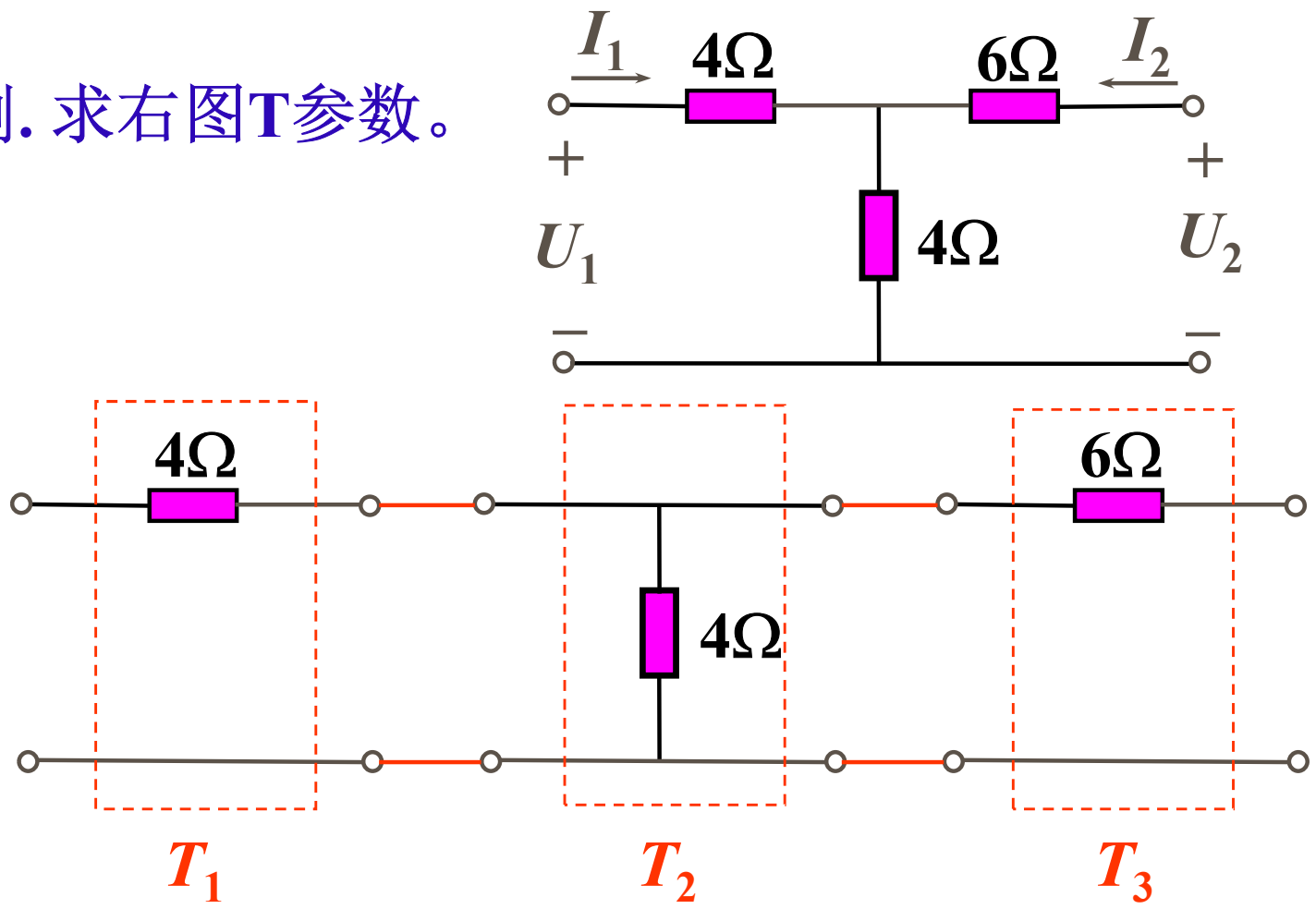
$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A' & B' \\ C' & D' \end{bmatrix} \begin{bmatrix} A'' & B'' \\ C'' & D'' \end{bmatrix}$$

$$= \begin{bmatrix} A'A'' + B'C'' & A'B'' + B'D'' \\ C'A'' + D'C'' & C'B'' + D'D'' \end{bmatrix}$$

显然 $A = A'A'' + B'C'' \neq A'A''$

(2) 级联时各双口网络的端口条件不会被破坏。

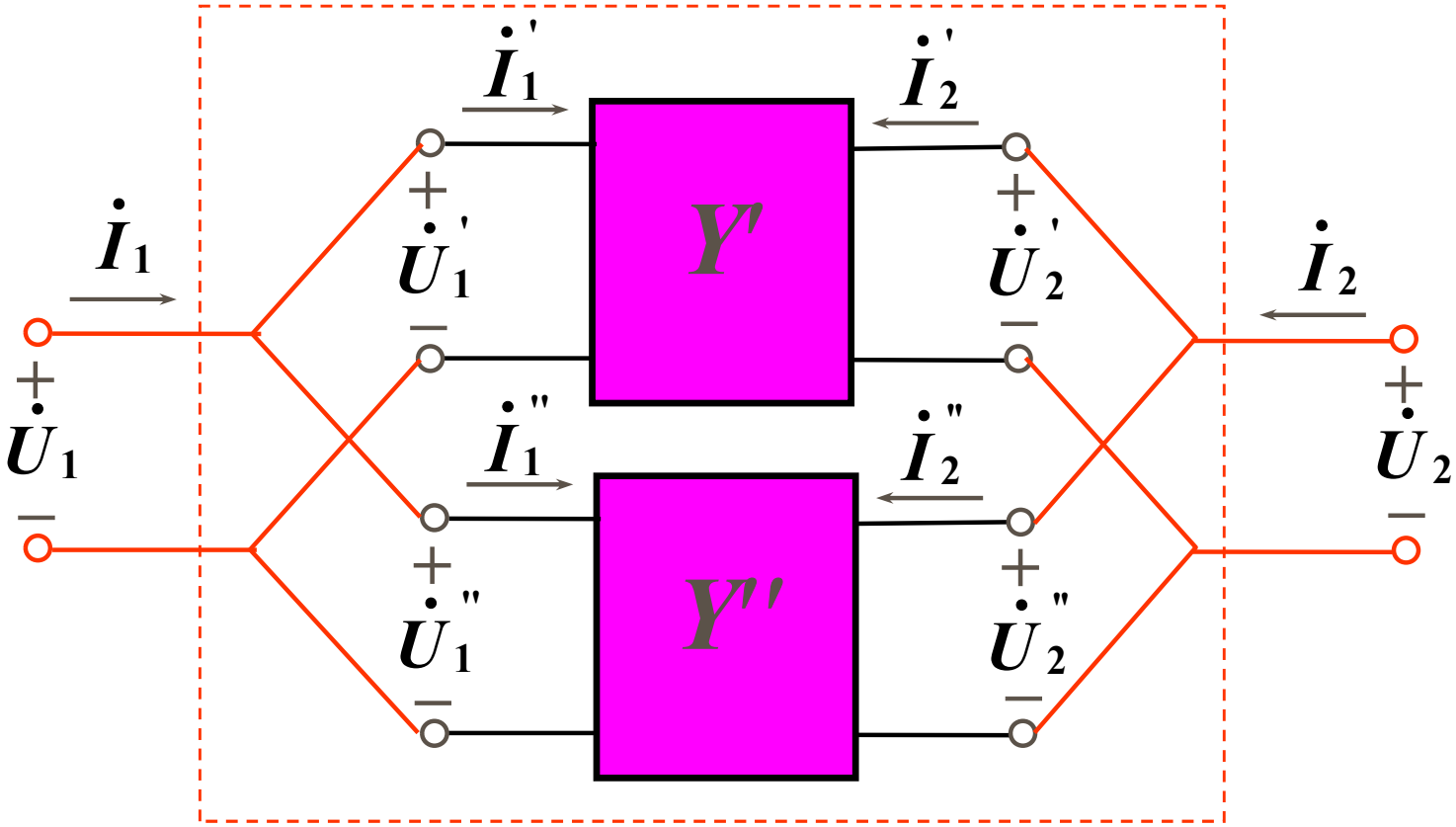
例. 求右图T参数。



易求出 $T_1=\begin{bmatrix}1 & 4\Omega\\ 0 & 1\end{bmatrix}$ $T_2=\begin{bmatrix}1 & 0\\ 0.25\,\mathrm{S} & 1\end{bmatrix}$ $T_3=\begin{bmatrix}1 & 6\Omega\\ 0 & 1\end{bmatrix}$

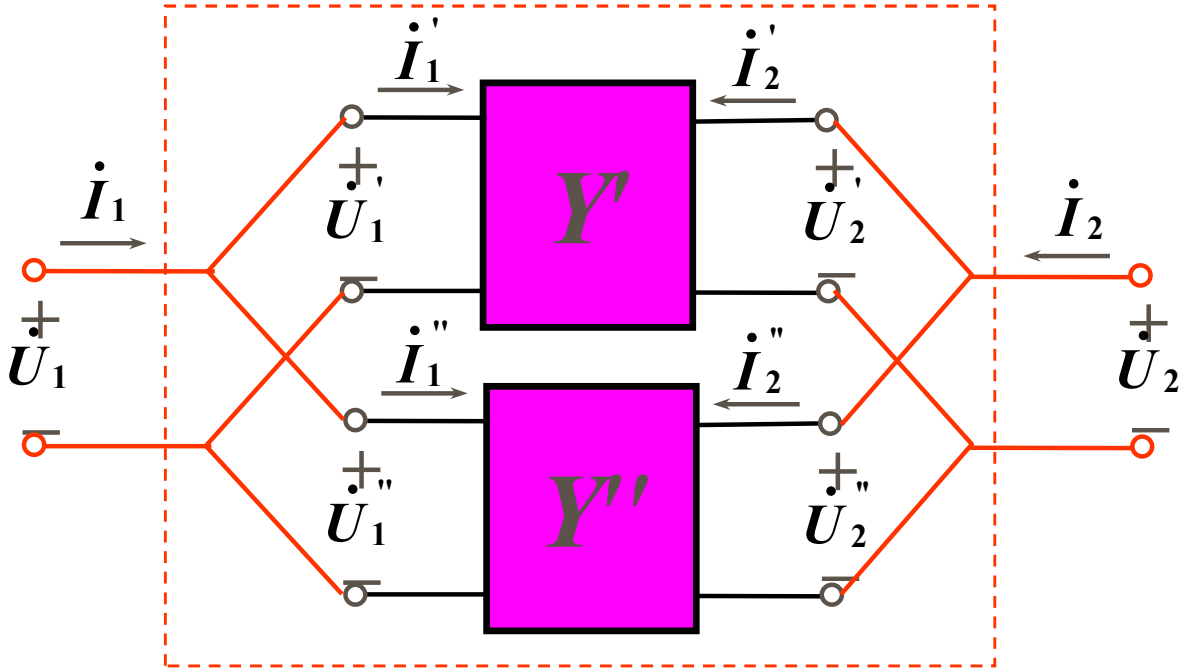
则 $[T]=[T_1][T_2][T_3]=\begin{bmatrix}1 & 4\\ 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0\\ 0.25 & 1\end{bmatrix}\begin{bmatrix}1 & 6\\ 0 & 1\end{bmatrix}=\begin{bmatrix}2 & 16\Omega\\ 0.25\,\mathrm{S} & 2.5\end{bmatrix}$

2、并联：联接方式如下图，并联采用Y参数方便。



$$\begin{bmatrix}\dot{I}_1'\\ \dot{I}_2'\end{bmatrix}=[Y']\begin{bmatrix}\dot{U}_1'\\ \dot{U}_2'\end{bmatrix}=\begin{bmatrix}Y_{11}' & Y_{12}'\\ Y_{21}' & Y_{22}'\end{bmatrix}\begin{bmatrix}\dot{U}_1'\\ \dot{U}_2'\end{bmatrix}$$

$$\begin{bmatrix}\dot{I}_1''\\ \dot{I}_2''\end{bmatrix}=[Y'']\begin{bmatrix}\dot{U}_1''\\ \dot{U}_2''\end{bmatrix}=\begin{bmatrix}Y_{11}'' & Y_{12}''\\ Y_{21}'' & Y_{22}''\end{bmatrix}\begin{bmatrix}\dot{U}_1''\\ \dot{U}_2''\end{bmatrix}$$



并联后

$$\begin{bmatrix}\dot{U}_1\\ \dot{U}_2\end{bmatrix}=\begin{bmatrix}\dot{U}_1'\\ \dot{U}_2'\end{bmatrix}=\begin{bmatrix}\dot{U}_1''\\ \dot{U}_2''\end{bmatrix}\qquad \begin{bmatrix}\dot{I}_1\\ \dot{I}_2\end{bmatrix}=\begin{bmatrix}\dot{I}_1'\\ \dot{I}_2'\end{bmatrix}+\begin{bmatrix}\dot{I}_1''\\ \dot{I}_2''\end{bmatrix}$$

$$\begin{bmatrix}\dot{I}_1\\ \dot{I}_2\end{bmatrix}=\begin{bmatrix}\dot{I}_1'\\ \dot{I}_2'\end{bmatrix}+\begin{bmatrix}\dot{I}_1''\\ \dot{I}_2''\end{bmatrix}=\begin{bmatrix}Y_{11}' & Y_{12}'\\ Y_{21}' & Y_{22}'\end{bmatrix}\begin{bmatrix}\dot{U}_1'\\ \dot{U}_2'\end{bmatrix}+\begin{bmatrix}Y_{11}'' & Y_{12}''\\ Y_{21}'' & Y_{22}''\end{bmatrix}\begin{bmatrix}\dot{U}_1''\\ \dot{U}_2''\end{bmatrix}$$

$$=\left\{\begin{bmatrix}Y_{11}' & Y_{12}'\\ Y_{21}' & Y_{22}'\end{bmatrix}+\begin{bmatrix}Y_{11}'' & Y_{12}''\\ Y_{21}'' & Y_{22}''\end{bmatrix}\right\}\begin{bmatrix}\dot{U}_1\\ \dot{U}_2\end{bmatrix}=\begin{bmatrix}Y_{11}'+Y_{11}'' & Y_{12}'+Y_{12}''\\ Y_{21}'+Y_{21}'' & Y_{22}'+Y_{22}''\end{bmatrix}\begin{bmatrix}\dot{U}_1\\ \dot{U}_2\end{bmatrix}$$

$$\begin{bmatrix}\dot{I}_1\\ \dot{I}_2\end{bmatrix}=\begin{bmatrix}Y_{11} & Y_{12}\\ Y_{11} & Y_{22}\end{bmatrix}\begin{bmatrix}\dot{U}_1\\ \dot{U}_2\end{bmatrix}=[Y]\begin{bmatrix}\dot{U}_1\\ \dot{U}_2\end{bmatrix}$$

可得 $[Y]=[Y']+[Y'']$

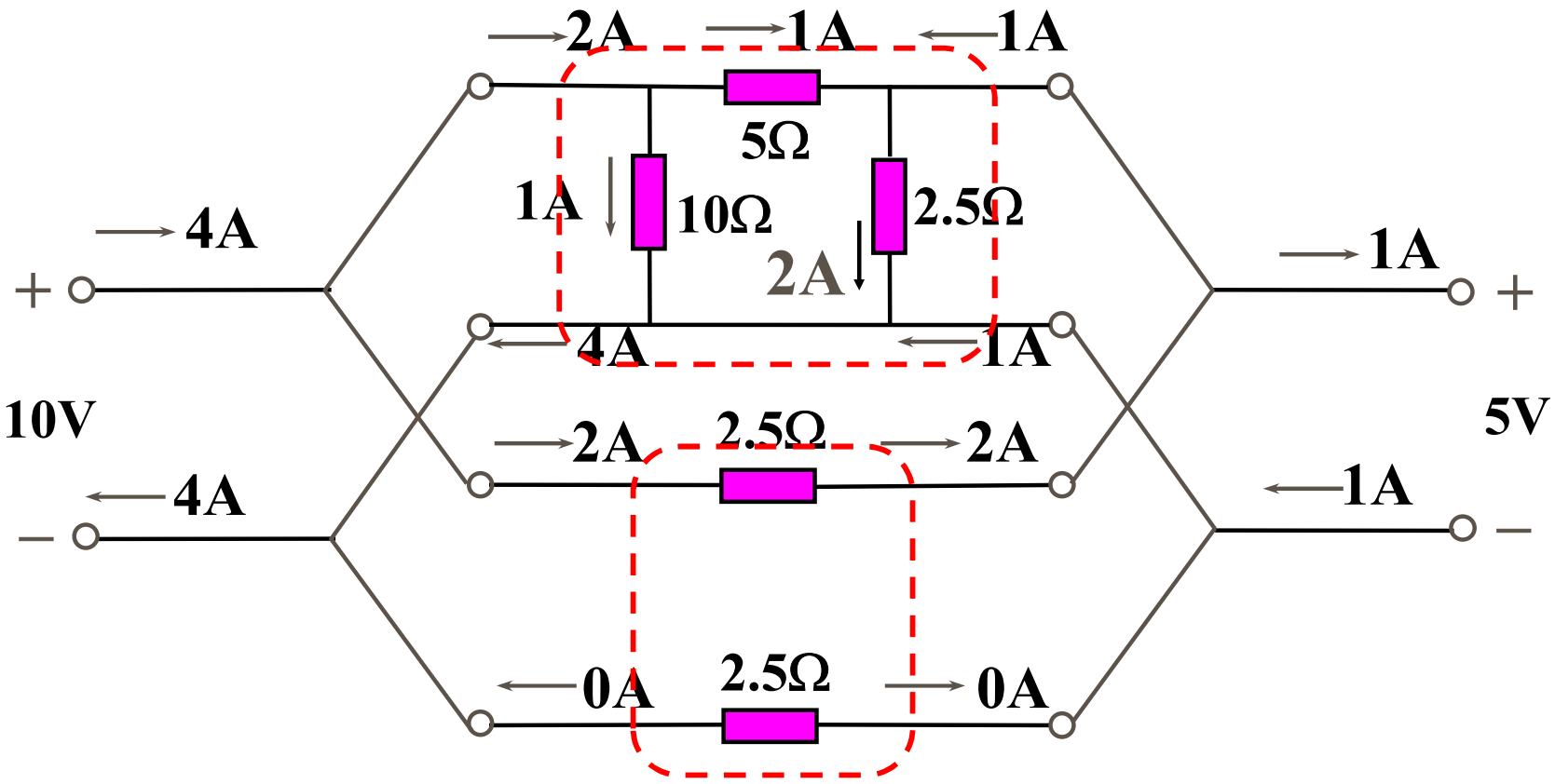
即 $$\begin{bmatrix}Y_{11} & Y_{12}\\ Y_{21} & Y_{22}\end{bmatrix}=\begin{bmatrix}Y'_{11} & Y'_{12}\\ Y'_{21} & Y'_{22}\end{bmatrix}+\begin{bmatrix}Y''_{11} & Y''_{12}\\ Y''_{21} & Y''_{22}\end{bmatrix}$$

结论：

双口网络并联所得复合双口网络的Y参数矩阵等于两个双口网络Y参数矩阵相加。

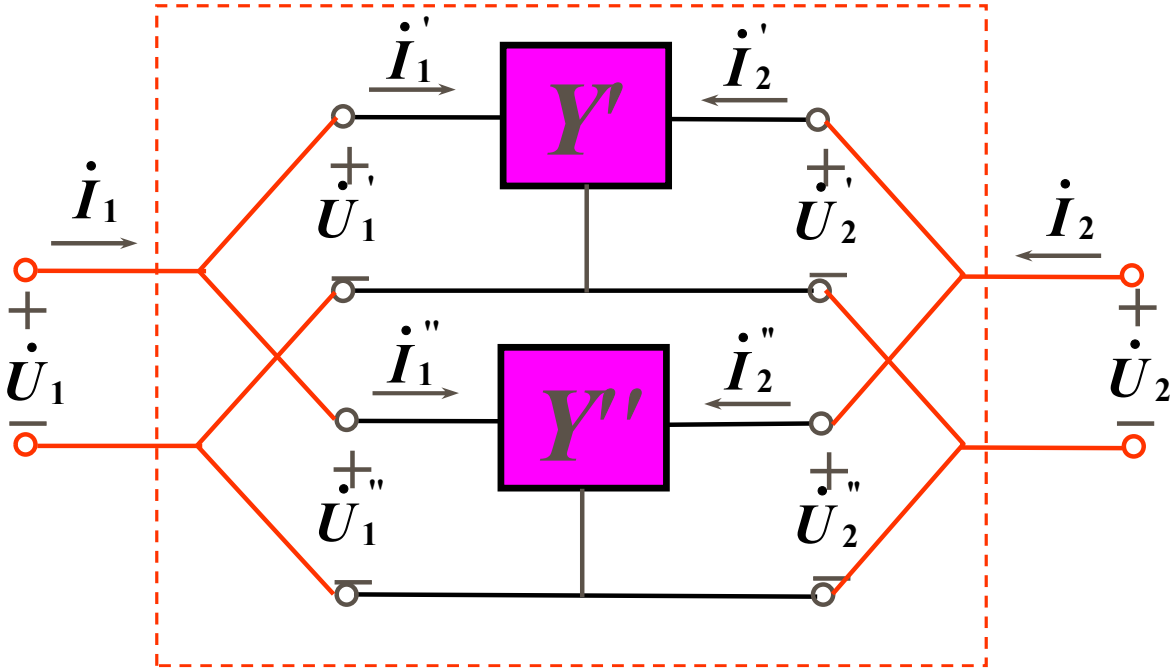
注意：

(1) 两个双口并联时，其端口条件可能被破坏此时上述关系式就不成立。

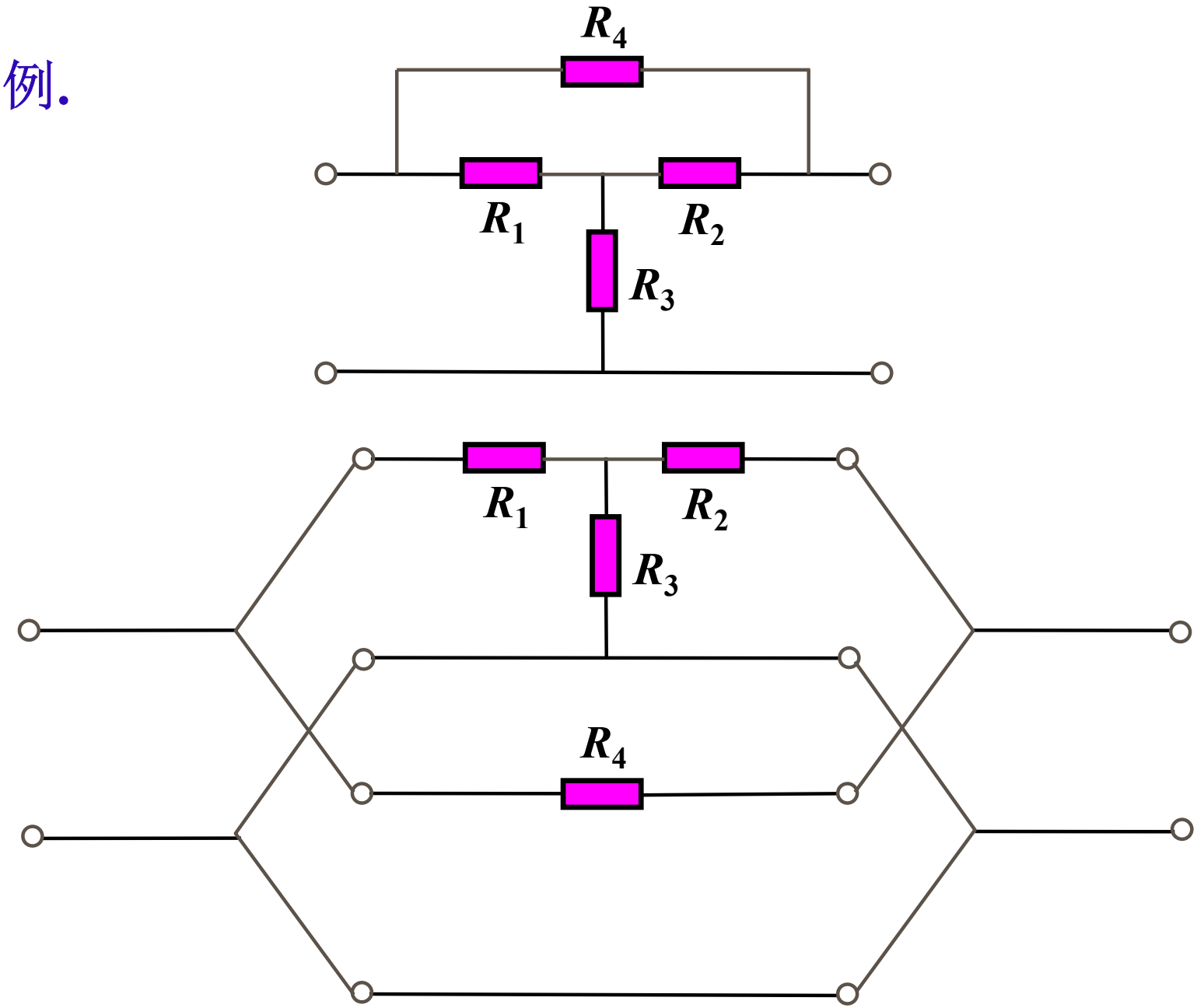


并联后端口条件破坏。

(2) 具有公共端的双口网络(三端网络形成的二端口)，将公共端并在一起将不会破坏端口条件。



例.

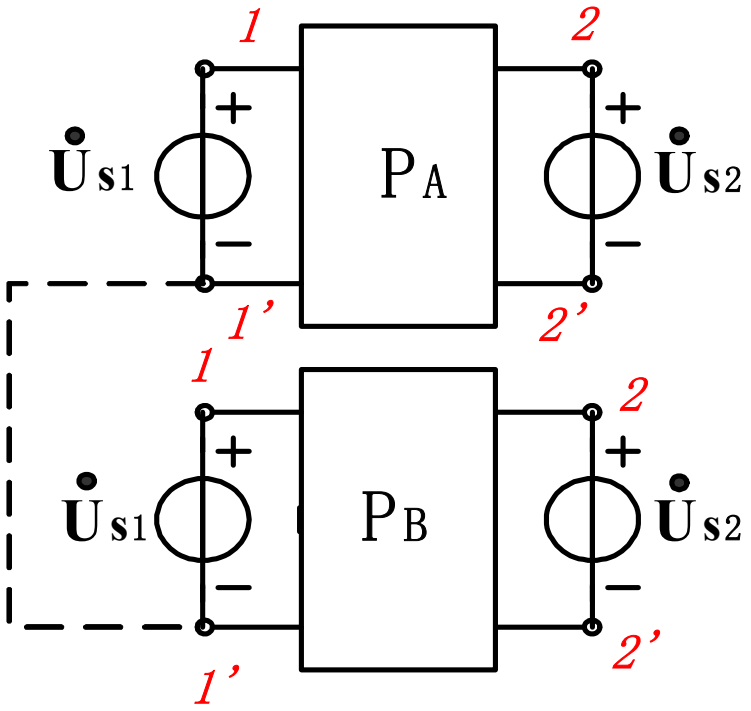


并联连接有效性判别方法：

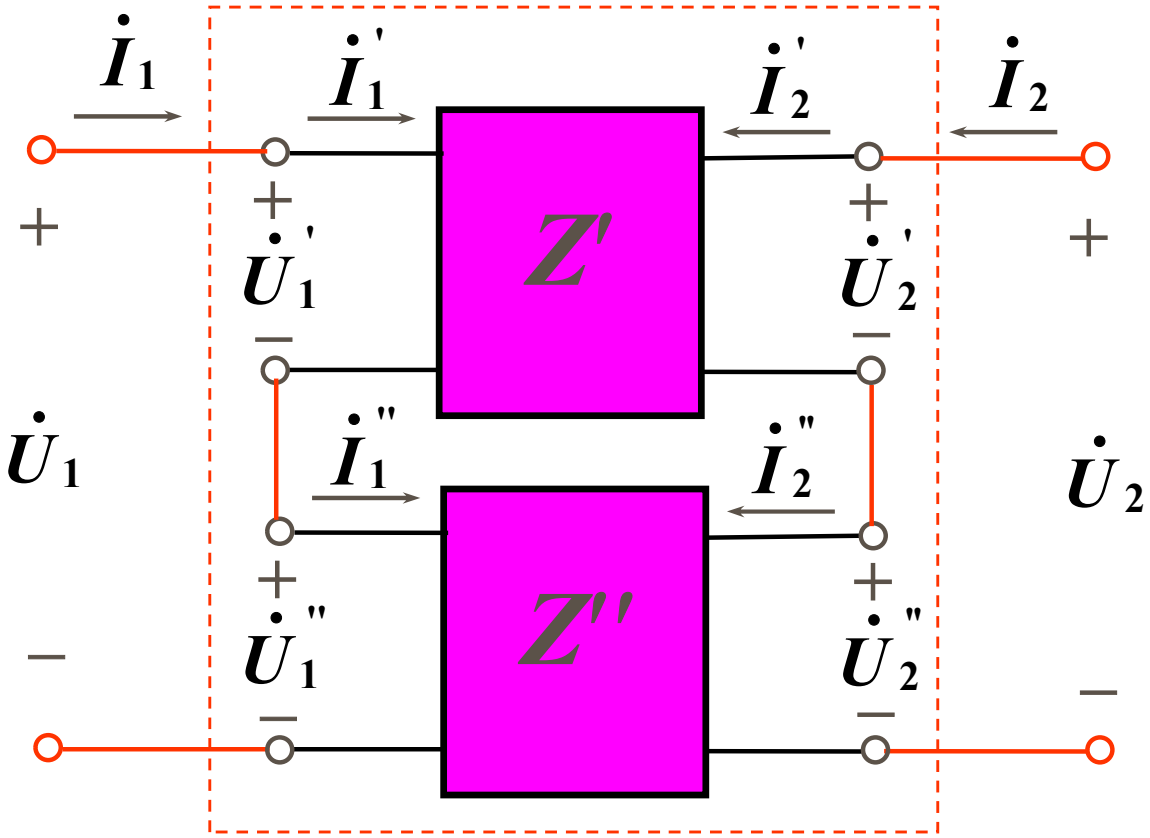
电路连接如图，对P_A、P_B二端分别加电压源$\dot{U}_{s1}$和$\dot{U}_{s2}$，计算或测量$\dot{U}_{A12}$ 和 $\dot{U}_{B12}$，若 $\dot{U}_{A12}=\dot{U}_{B12}$ 则满足并联有效性，可用并联计算式来计算合成后短路参数。

$$\dot{U}_{A12}=\dot{U}_{B12} \rightarrow \dot{U}_{A1'2'}=\dot{U}_{B1'2'}$$

两个网络中**1,2,2'**为等位点，并联连接后不改变原网络各支路电压电流。原端口电流不变。

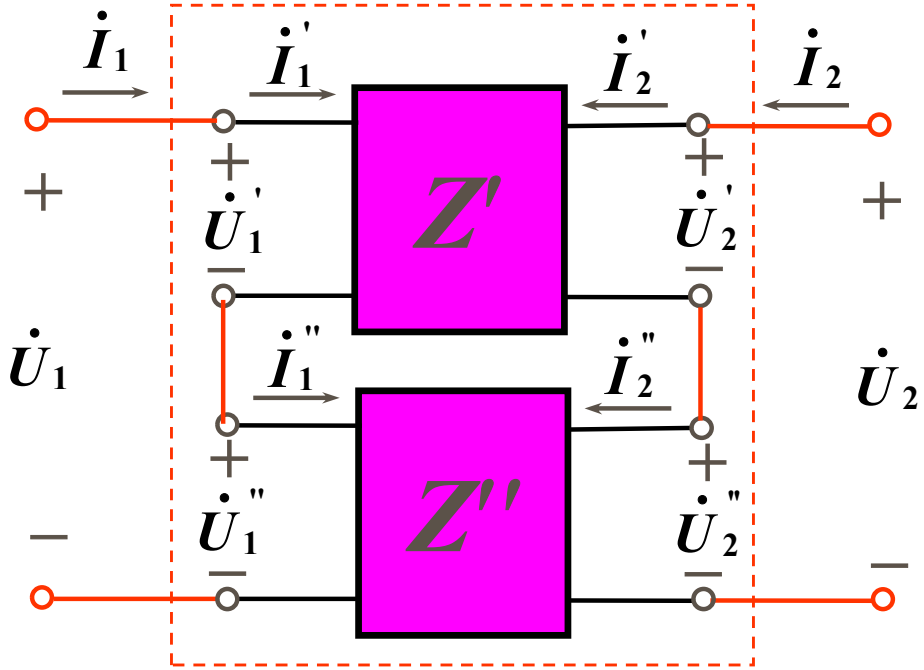


3、串联：联接方式如图，采用Z 参数方便。



$$\begin{bmatrix}\dot{U}_1' \\ \dot{U}_2'\end{bmatrix} = [Z'] \begin{bmatrix}\dot{I}_1' \\ \dot{I}_2'\end{bmatrix} = \begin{bmatrix}Z_{11}' & Z_{12}' \\ Z_{21}' & Z_{22}'\end{bmatrix} \begin{bmatrix}\dot{I}_1' \\ \dot{I}_2'\end{bmatrix}$$

$$\begin{bmatrix}\dot{U}_1'' \\ \dot{U}_2''\end{bmatrix} = [Z''] \begin{bmatrix}\dot{I}_1'' \\ \dot{I}_2''\end{bmatrix} = \begin{bmatrix}Z_{11}'' & Z_{12}'' \\ Z_{21}'' & Z_{22}''\end{bmatrix} \begin{bmatrix}\dot{I}_1'' \\ \dot{I}_2''\end{bmatrix}$$



$$\begin{bmatrix}\dot{I}_1\\ \dot{I}_2\end{bmatrix}=\begin{bmatrix}\dot{I}_1'\\ \dot{I}_2'\end{bmatrix}=\begin{bmatrix}\dot{I}_1''\\ \dot{I}_2''\end{bmatrix}\qquad \begin{bmatrix}\dot{U}_1\\ \dot{U}_2\end{bmatrix}=\begin{bmatrix}\dot{U}_1'\\ \dot{U}_2'\end{bmatrix}+\begin{bmatrix}\dot{U}_1''\\ \dot{U}_2''\end{bmatrix}$$

$$\begin{bmatrix}\dot{U}_1\\ \dot{U}_2\end{bmatrix}=\begin{bmatrix}\dot{U}_1'\\ \dot{U}_2'\end{bmatrix}+\begin{bmatrix}\dot{U}_1''\\ \dot{U}_2''\end{bmatrix}=[Z']\begin{bmatrix}\dot{I}_1'\\ \dot{I}_2'\end{bmatrix}+[Z'']\begin{bmatrix}\dot{I}_1''\\ \dot{I}_2''\end{bmatrix}$$

$$=\{[Z']+[Z'']\}\begin{bmatrix}\dot{I}_1\\ \dot{I}_2\end{bmatrix}=[Z]\begin{bmatrix}\dot{I}_1\\ \dot{I}_2\end{bmatrix}$$

则 $$[Z]=[Z']+[Z'']$$

即 $$\begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix}=\begin{bmatrix} Z'_{11} & Z'_{12} \\ Z'_{21} & Z'_{22} \end{bmatrix}+\begin{bmatrix} Z''_{11} & Z''_{12} \\ Z''_{21} & Z''_{22} \end{bmatrix}$$

结论：

串联后复合双口Z 参数矩阵等于原双口Z 参数矩阵相加。可推广到n端口串联。

注意：

(1) 串联后端口条件可能被破坏。需检查端口条件。

(2) 具有公共端的二端口，将公共端串联时将不会破坏端口条件。

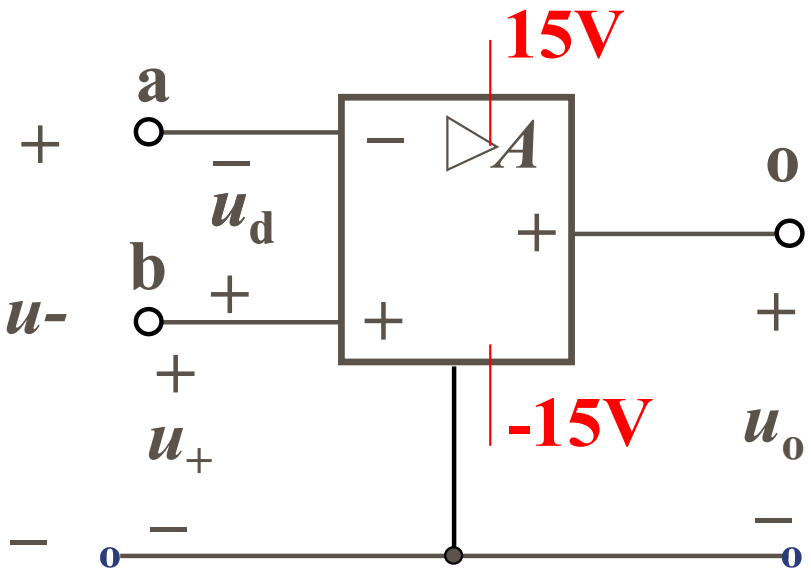
10.9 运算放大器

运算放大器(**operational amplifier**)：是高放大倍数的放大器



放大倍数 $A=\dfrac{u_{sc}}{u_{sr}}=10^6$

1. 电路符号



a：反向输入端，输入电压 u_-

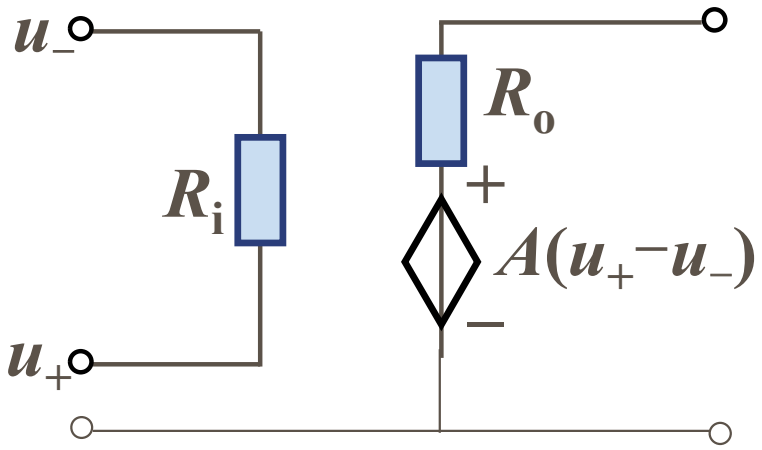
b：同向输入端，输入电压 u_+

o：输出端，输出电压 u_o

A：开环电压放大倍数，可达几百万倍

运算放大器需要电源供电。

2. 电路模型及特点



R_i：运算放大器两输入端间的输入电阻。10^6~$10^8\Omega$

R_o：运算放大器的输出电阻。100 Ω

特点：

高放大倍数 (A:几万~几百万倍)
高输入电阻 (R_i: 几兆~几十兆)
低输出电阻 (R_o: 几十欧)

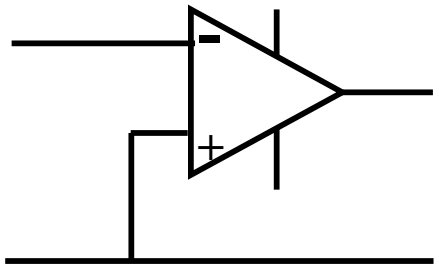
理想化 → 近似

$A = \infty$

$R_i = \infty$

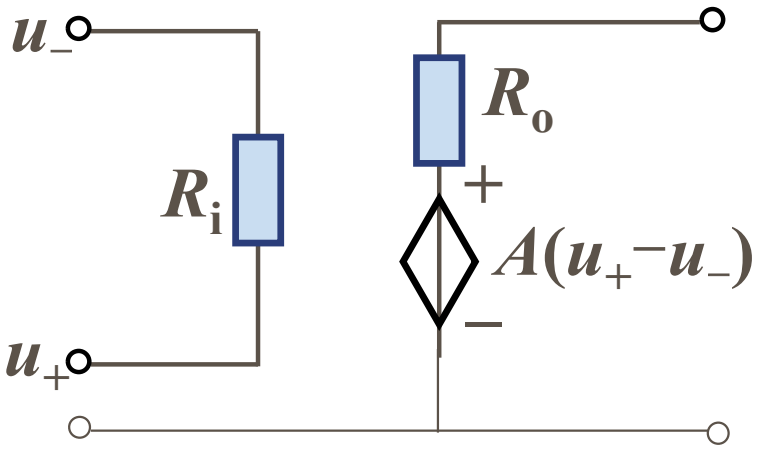
$R_o = 0$

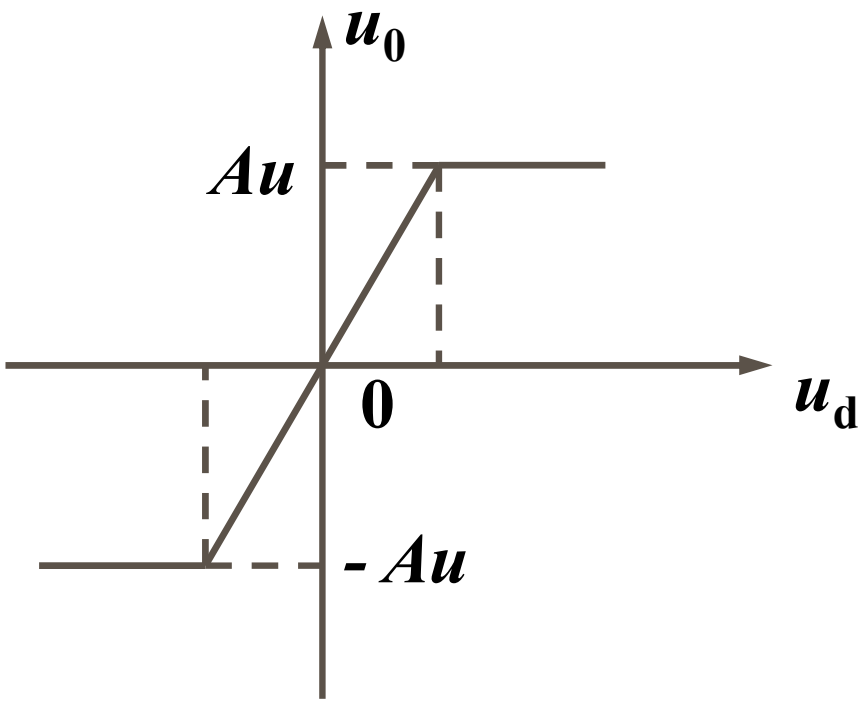
$R_i \approx \infty$　输入电流近似为0，$i \approx 0$　虚断



运放常用电路符号

2. 输出输入特性





$R_0 \approx 0$ $\quad u_0 = A(u_+ - u_-)$

$A \approx \infty$ $\quad u_+ - u_- \approx 0$

$u_+ \approx u_-$ 虚短

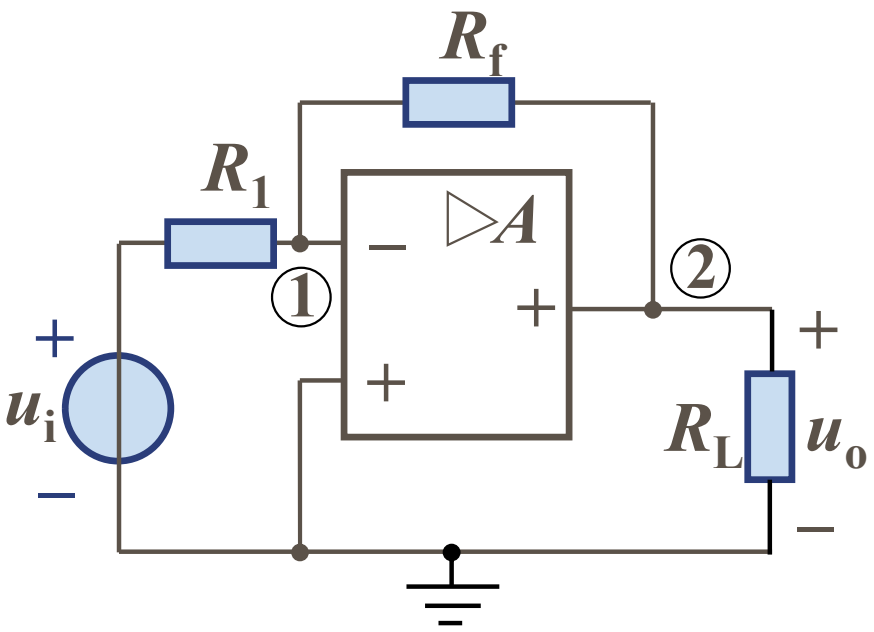
$i \approx 0$ 虚断

$u_+ \approx u_-$ 虚短

含运算放大器的电路的分析

运放开环工作极不稳定，一般外部接若干元件(*R*、*C*等)，使其工作在闭环状态。

1. 反相比例器



“虚短”： $u_+ = u_- = 0$，

$i_1 = u_S/R_1 \quad i_2 = -u_o/R_f$

“虚断”： $i_- = 0$，$i_2 = i_1$

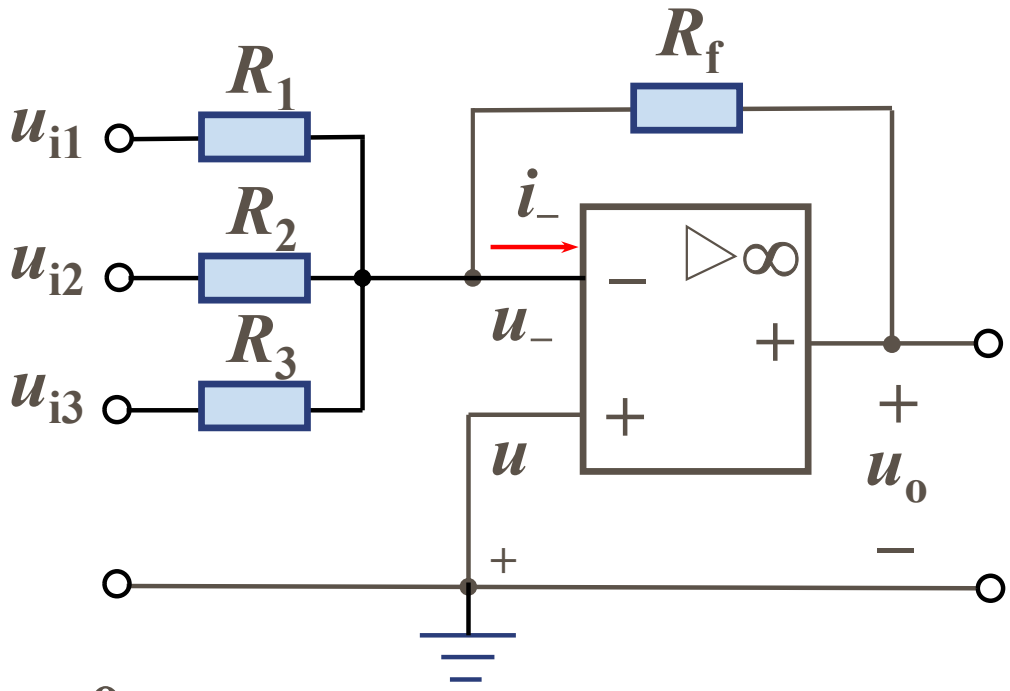
$$\therefore \quad u_o = -\frac{R_f}{R_1} u_i$$

输入电压放大倍数 $\frac{R_f}{R_1}$，

相位反相。

相当于一个压控电压源。

2. 加法器

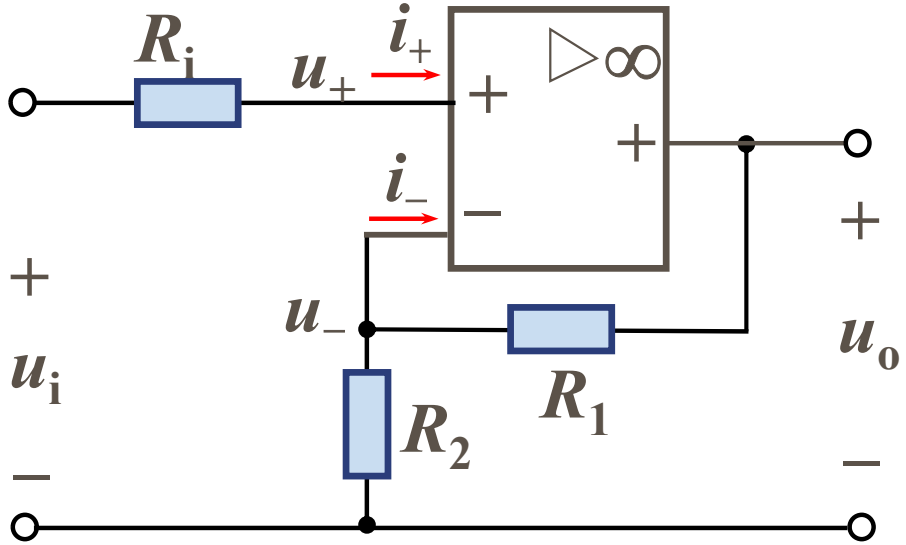


$$\begin{cases} u_- = u_+ = 0 \\ i_- = 0 \end{cases}$$

$$u_{i1}/R_1 + u_{i2}/R_2 + u_{i3}/R_3 = -u_o/R_f$$

$$u_o = -(R_f/R_1\, u_{i1} + R_f/R_2\, u_{i2} + R_f/R_3\, u_{i3})$$

3. 正相比例器



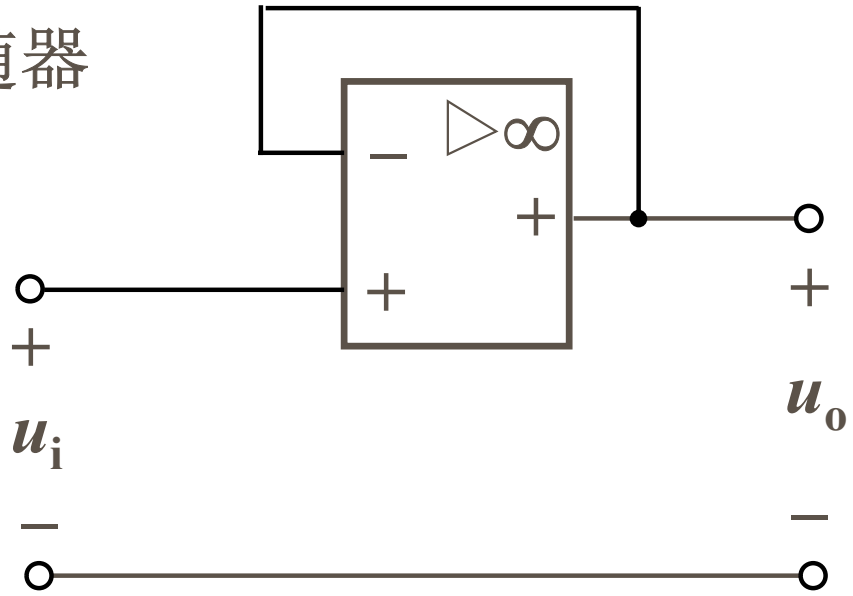
$$\begin{cases} u_+ = u_- = u_i \\ i_+ = i_- = 0 \end{cases}$$

$$(u_o - u_-)/R_1 = u_-/R_2$$

$$u_o = (R_1 + R_2)/R_2 \; u_i$$

$$= (1 + R_1/R_2) \; u_i$$

4. 电压跟随器



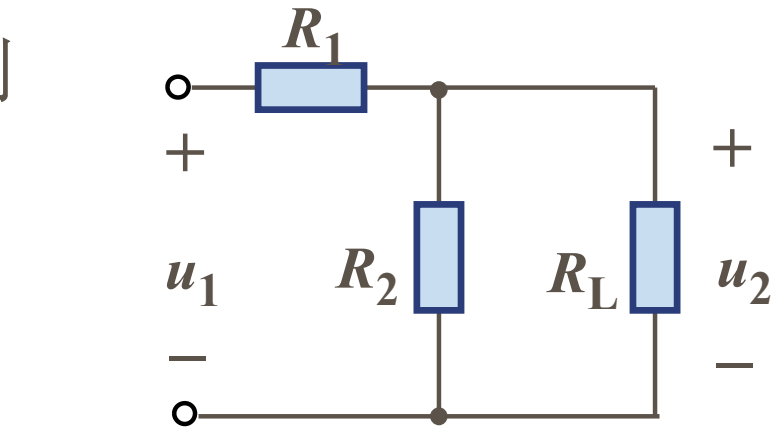
特点：① 输入阻抗无穷大(虚断)；

② 输出阻抗为零；

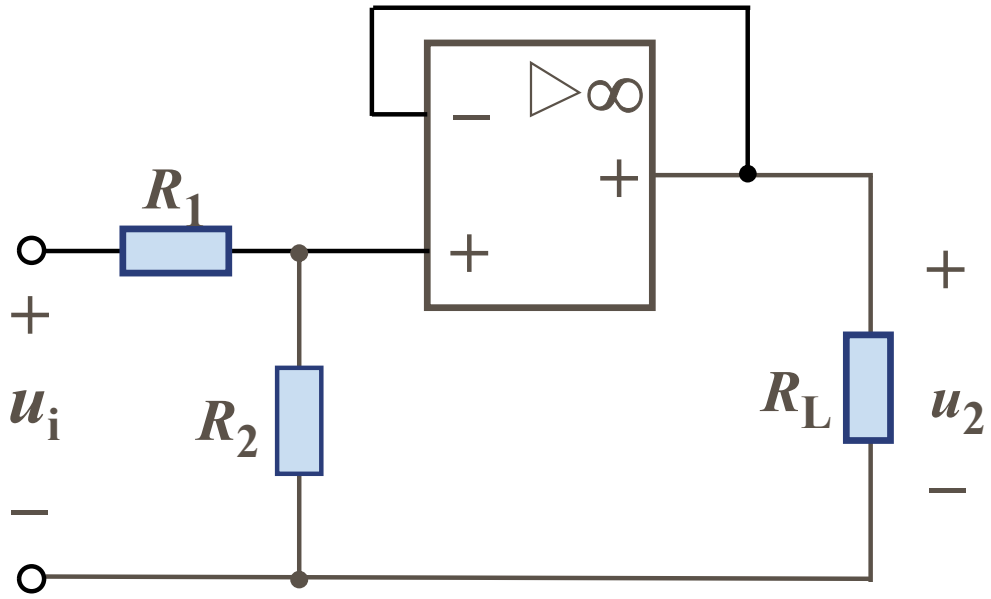
③ $u_o = u_i$。

应用：在电路中起隔离前后两级电路的作用。

例



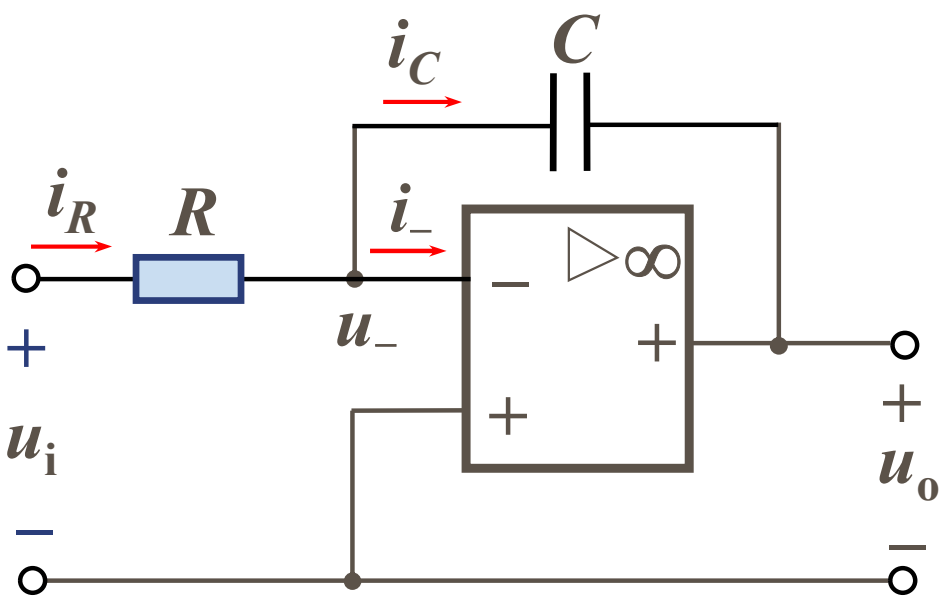
$$u_2 \neq \frac{R_2}{R_1 + R_2} u_1$$



$$u_2 = \frac{R_2}{R_1 + R_2} u_1$$

可见，加入跟随器后，隔离了前后两级电路的相互影响。

5. 积分器



$$\begin{cases} u_-=0 \\ i_-=0 \end{cases}$$

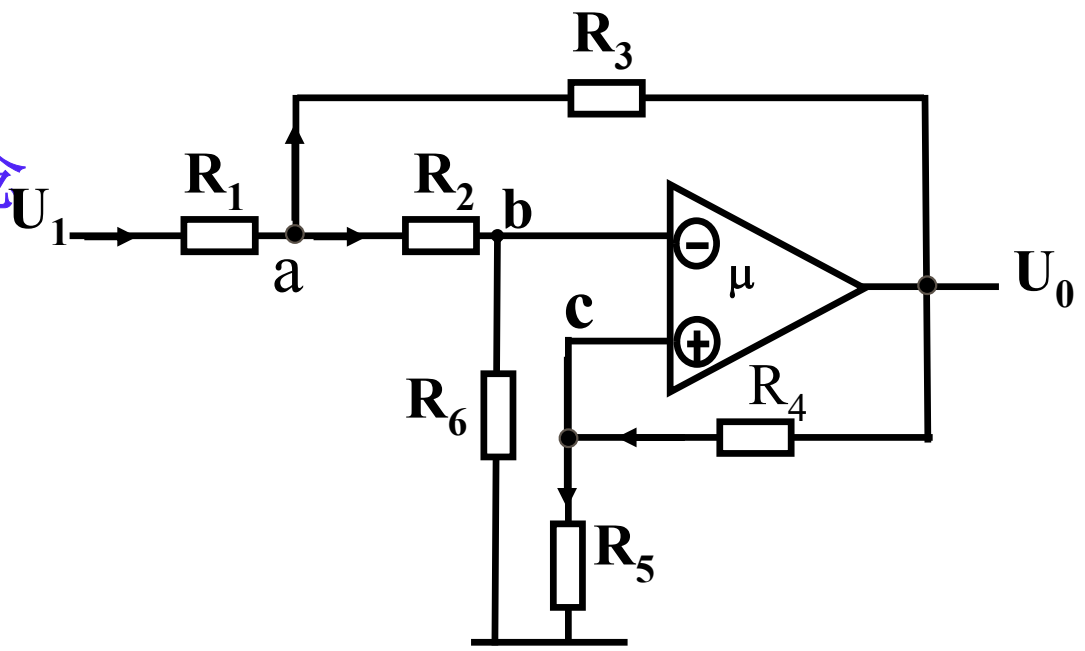
$$i_R = i_C$$

$$\frac{u_i}{R} = -C\frac{\mathrm{d}u_O}{\mathrm{d}t}$$

$$u_0 = -\frac{1}{RC}\int u_i \mathrm{d}t$$

返回首页

例： 如图所示电路，求$\frac{U_0}{U_1}$。



解一：利用“虚短、虚断”概念

$$a:\frac{U_1-U_a}{R_1}=\frac{U_a-U_b}{R_2}+\frac{U_a-U_0}{R_3}$$

$$b:\frac{U_a-U_b}{R_2}=\frac{U_b}{R_6}$$

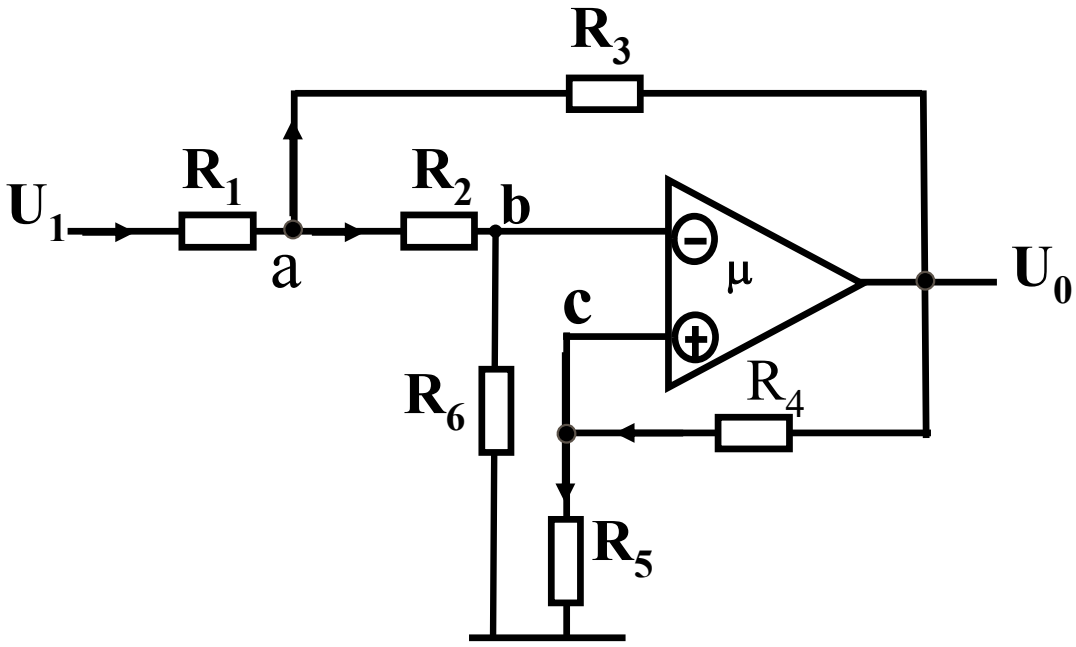
$$c:\frac{U_0-U_c}{R_4}=\frac{U_c}{R_5}$$

$U_b=U_c$ 虚短

消去U_a、U_b、U_c，得

$$\frac{U_0}{U_1}=\frac{R_3R_6(R_4+R_5)}{R_5(R_1R_2+R_2R_3+R_3R_1+R_3R_6)-R_1R_4R_6}$$

解二：由节点电压方程得



$$\text{a: } U_a\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right)-$$

$$U_b\frac{1}{R_2}-U_0\frac{1}{R_3}=U_1\frac{1}{R_1}$$

$$\mathbf{b:}\ \mathbf{U_b\left(\frac{1}{R_2}+\frac{1}{R_6}\right)-U_a\frac{1}{R_2}=0}$$

$$\mathbf{c:}\ \mathbf{U_c\left(\frac{1}{R_4}+\frac{1}{R_5}\right)-U_0\frac{1}{R_4}=0}$$

$$U_b=U_c$$

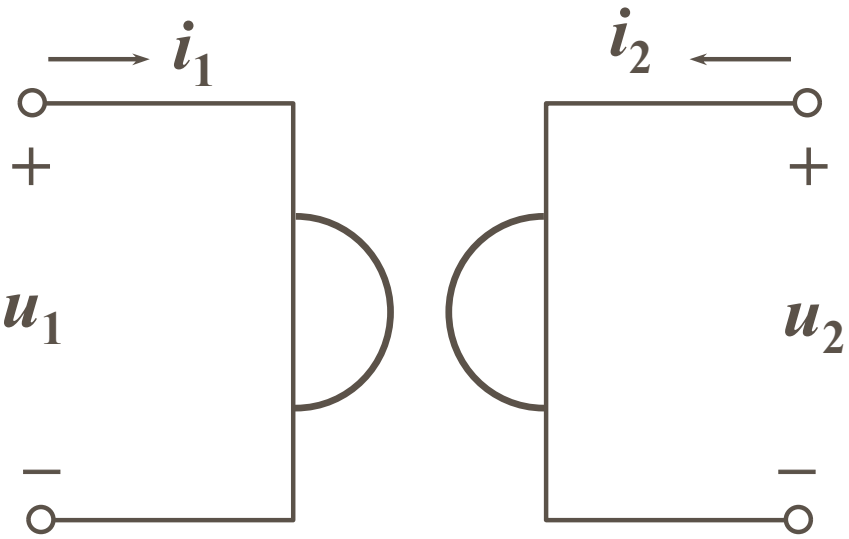
由以上四式，可解得：

$$\frac{U_0}{U_1}=\frac{R_3R_6(R_4+R_5)}{R_5(R_1R_2+R_2R_3+R_3R_1+R_3R_6)-R_1R_4R_6}$$

8.10 回转器

（1） 回转器：回转器也是二端口.

电路符号



特性：

$$\begin{cases} u_1 = -ri_2 \\ u_2 = ri_1 \end{cases}$$ 或 $$\begin{cases} i_1 = gu_2 \\ i_2 = -gu_1 \end{cases}$$ $$\begin{cases} u_1 = -ri_2 \\ i_1 = \dfrac{1}{r}u_2 \end{cases}$$

r 称为回转电阻　　g 称为回转电导　　T参数方程

其矩阵形式为：

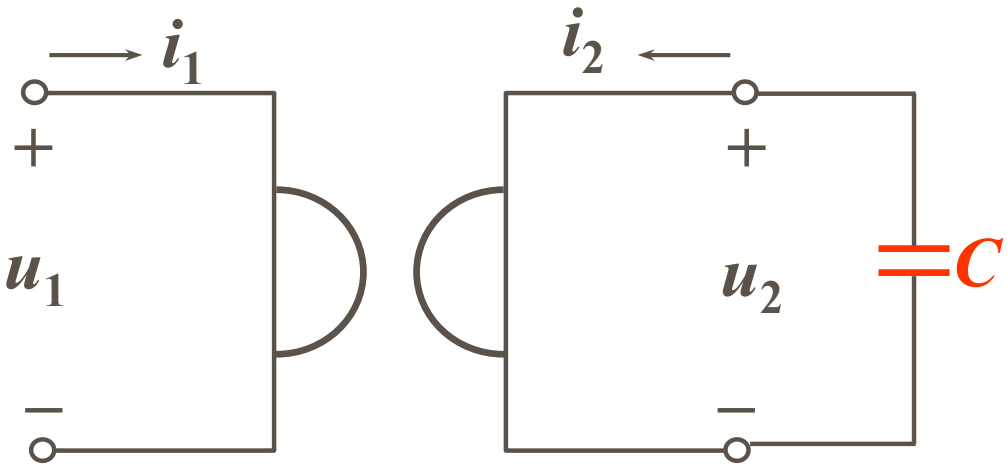
$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 & -r \\ r & 0 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} \quad \text{或} \quad \begin{bmatrix} i_1 \\ i_2 \end{bmatrix} = \begin{bmatrix} 0 & g \\ -g & 0 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

注意u, i的方向!

令 $\quad Z = \begin{bmatrix} 0 & -r \\ r & 0 \end{bmatrix} \qquad Y = \begin{bmatrix} 0 & g \\ -g & 0 \end{bmatrix}$

有 $\quad Z = Y^{-1}$

（2） 回转器可以把一个端口的电流(或电压)回转成另一个端口的电压(或电流)。利用回转器可以把电容回转成电感。



$$\begin{cases} u_1 = -ri_2 \\ u_2 = ri_1 \\ i_2 = -C\dfrac{du_2}{dt} \end{cases}$$

$$u_1 = -ri_2 = rC\frac{du_2}{dt}$$

$$= r^2C\frac{di_1}{dt} = L\frac{di_1}{dt}$$

$$\boxed{L = r^2C}$$

从端口1看，u_1, i_1关系为一等效电感关系，$L= r^2C$.

若　r =50kΩ, C =1μF

则　等效电感　L=2500H！

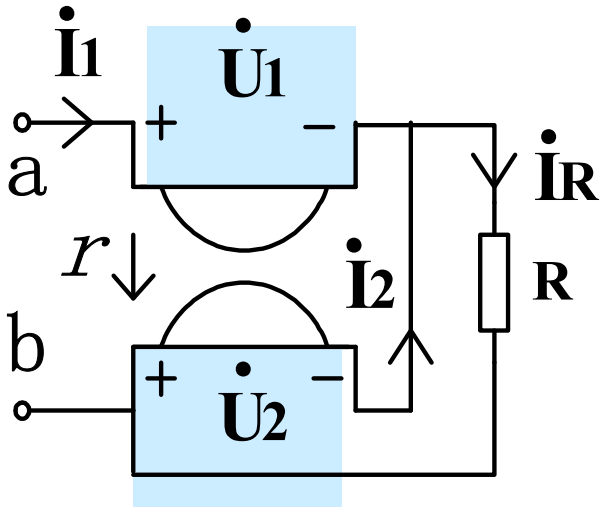
（3） 回转器不消耗功率(能量)，也不储能。是线性无源元件。

$$u_1 i_1 + u_2 i_2 = -r i_2 i_1 + r i_1 i_2 = 0$$

（4） 回转器是非互易元件。

$$Z = \begin{bmatrix} 0 & -r \\ r & 0 \end{bmatrix} \qquad Y = \begin{bmatrix} 0 & g \\ -g & 0 \end{bmatrix}$$

例: 求如图电路的入端阻抗



解:

$$Z_{ab}=\frac{\dot{U}_{ab}}{\dot{I}_1}\left(=\frac{\dot{U}_1-\dot{U}_2}{\dot{I}_1}=\frac{()\dot{I}_1}{\dot{I}_1}\right)$$

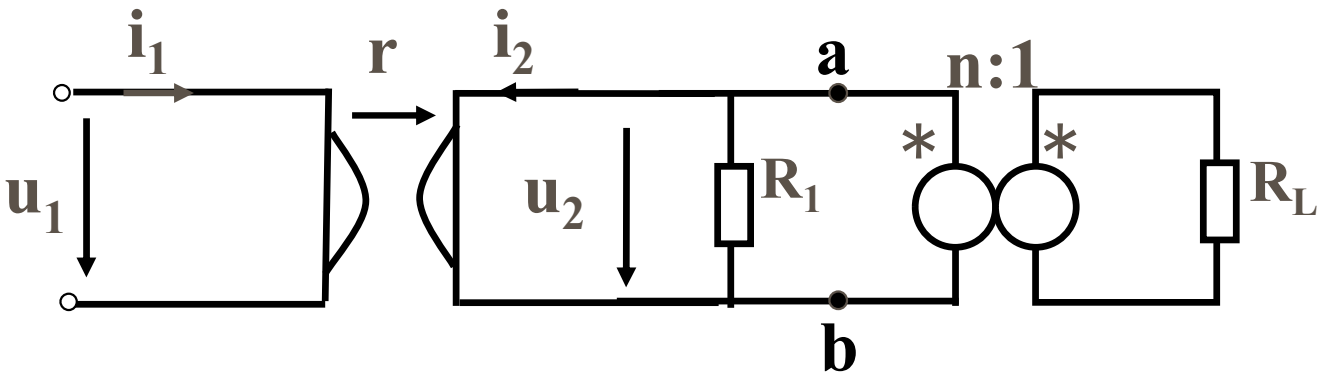
$$\dot{U}_2=r\dot{I}_1 \longrightarrow \dot{I}_R=-\frac{\dot{U}_2}{R}=-\frac{r\dot{I}_1}{R}$$

$$\dot{I}_2=\dot{I}_R-\dot{I}_1=-\frac{r\dot{I}_1}{R}-\dot{I}_1=-\frac{r+R}{R}\dot{I}_1 \longrightarrow \dot{U}_1=-r\dot{I}_2=r\frac{r+R}{R}\dot{I}_1$$

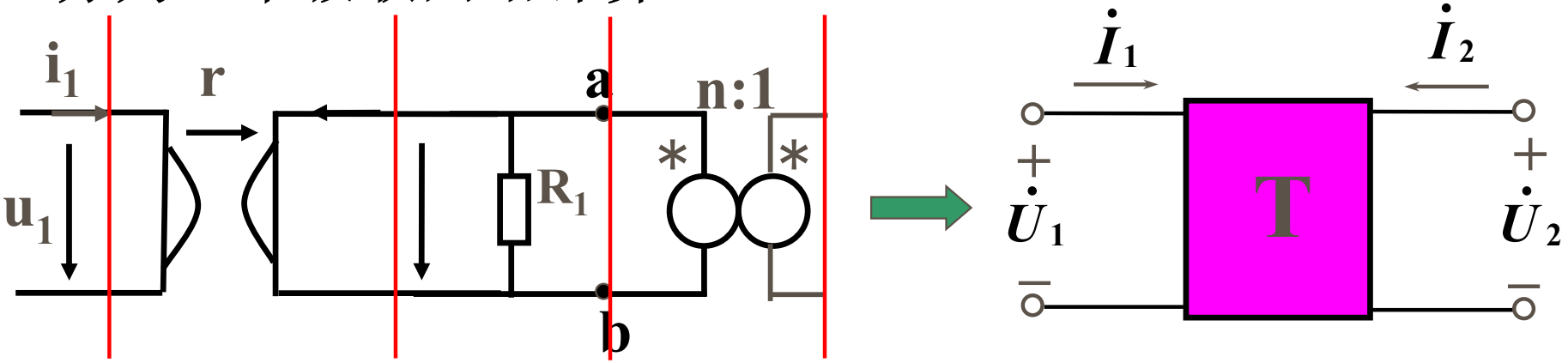
$$Z_{ab}=\frac{\dot{U}_{ab}}{\dot{I}_1}=\frac{\dot{U}_1-\dot{U}_2}{\dot{I}_1}=r\frac{r+R}{R}-r=\frac{r^2}{R}$$

$$\begin{cases}\dot{U}_1=-r\dot{I}_2\\ \dot{I}_1=\dfrac{1}{r}\dot{U}_2\end{cases}$$

例: 如图所示电路，已知$\dot{U}_1 = 10\angle 0^\circ\ \mathrm{V}$ ，$r=4\,\Omega$ ，$R_1 = 2\,\Omega$ ，$n = 2$ 。 求 $R_L = ?$时， R_L获得最大功率 $P_{max} = ?$

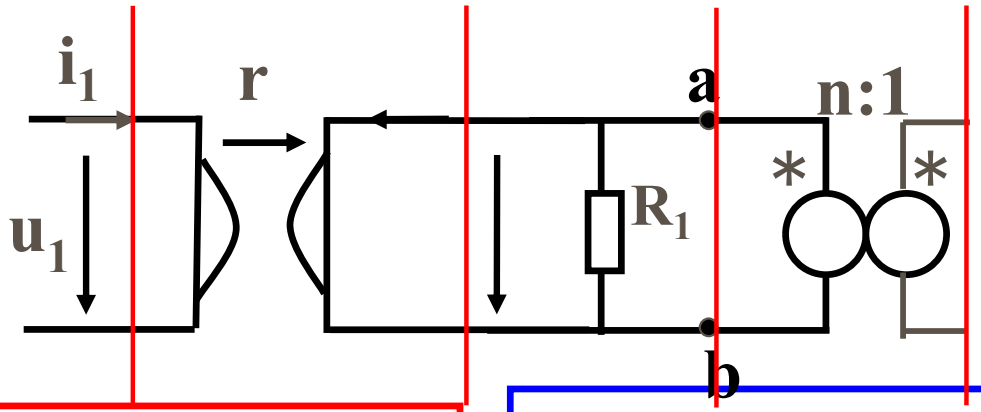


解1：分为三个级联网络计算



$$T_1=\begin{bmatrix}0 & r\\ \frac{1}{r} & 0\end{bmatrix}\qquad T_2=\begin{bmatrix}1 & 0\\ \frac{1}{R_1} & 1\end{bmatrix}$$

$$T_3=\begin{bmatrix}n & 0\\ 0 & \frac{1}{n}\end{bmatrix}$$



$$\frac{\dot U_1}{\dot U_2}=n\qquad \frac{\dot I_1}{\dot I_2}=-\frac{1}{n}$$

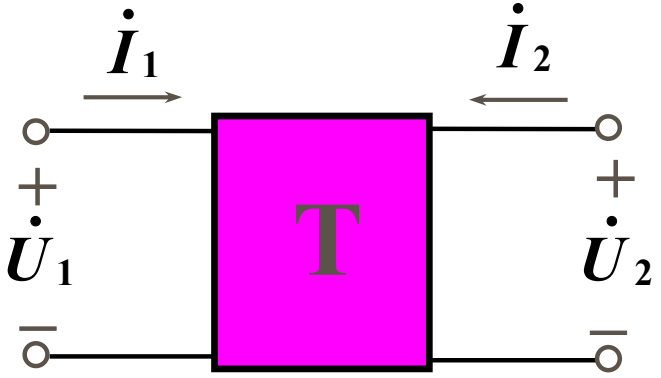
$$\begin{cases}\dot U_1=A\dot U_2+B(-\dot I_2)\\ \dot I_1=C\dot U_2+D(-\dot I_2)\end{cases}$$

$$T=[T_1][T_2][T_3]=\begin{bmatrix}0 & r\\ \frac{1}{r} & 0\end{bmatrix}\begin{bmatrix}1 & 0\\ \frac{1}{R_1} & 1\end{bmatrix}\begin{bmatrix}n & 0\\ 0 & \frac{1}{n}\end{bmatrix}=\begin{bmatrix}\frac{nr}{R_1} & \frac{r}{n}\\ \frac{n}{r} & 0\end{bmatrix}=\begin{bmatrix}4 & 2\\ \frac{1}{2} & 0\end{bmatrix}$$

$$\begin{cases}\dot U_1=4\dot U_2+2(-\dot I_2)\\ \dot I_1=\frac{1}{2}\dot U_2\end{cases}$$

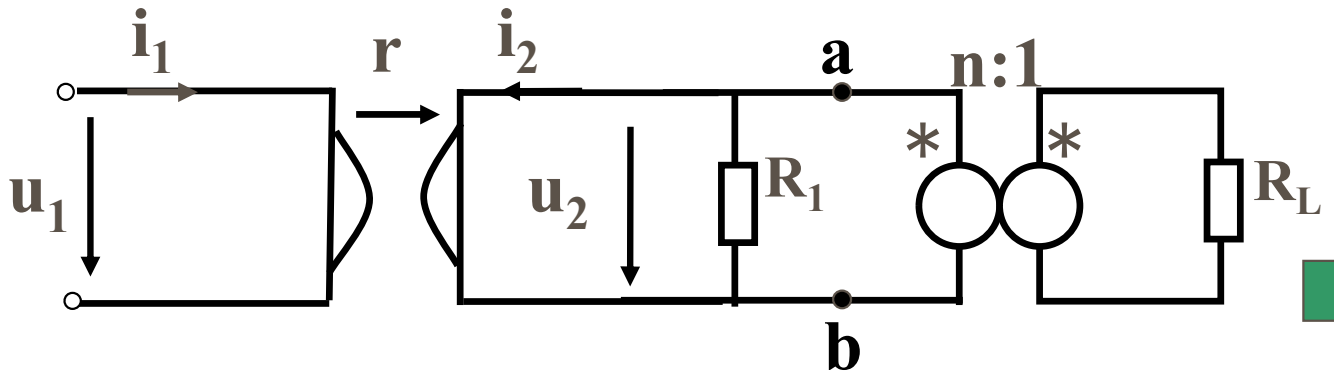
开路电压 $U_{20}\big|_{I_2=0}=\frac{5}{2}\angle 0^\circ$ V

短路电流 $-I_{2d}\big|_{U_2=0}=5\angle 0^\circ$ A

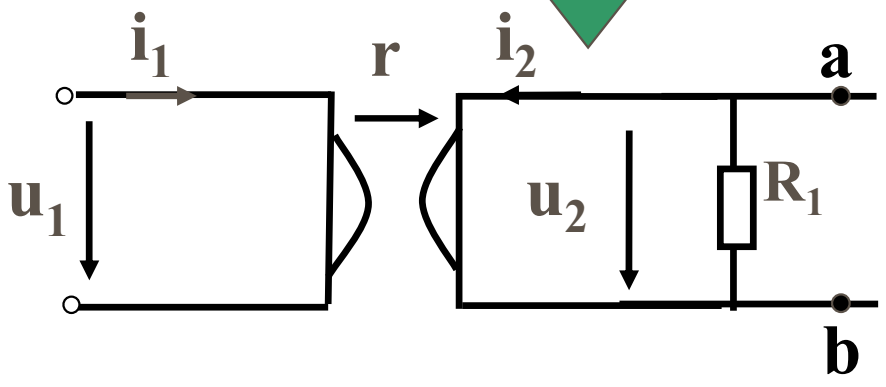


入端阻抗 $R_0=\frac{1}{2}\Omega=R_L$时，$P_{max}=\frac{U_{20}^2}{4R_0}=\frac{25}{8}\omega$

解2：可先求ab左端的戴维南等效电路，由题意知



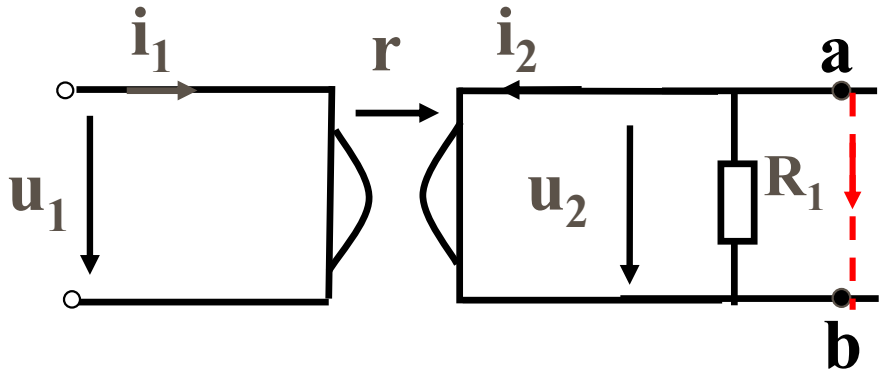
$$\begin{cases} \dot{U}_1 = -r\,\dot{I}_2 \\ \dot{I}_1 = \dfrac{1}{r}\dot{U}_2 \end{cases}$$



$$-\dot{I}_2 = \frac{\dot{U}_1}{r} = \frac{10\angle 0^\circ}{4} = \frac{5}{2}\angle 0^\circ \ \text{A}$$

开路电压 $\dot{U}_{20} = R_1 \cdot (-\dot{I}_2) = 2 \times \dfrac{5}{2}\angle 0^\circ = 5\angle 0^\circ \ \text{V}$

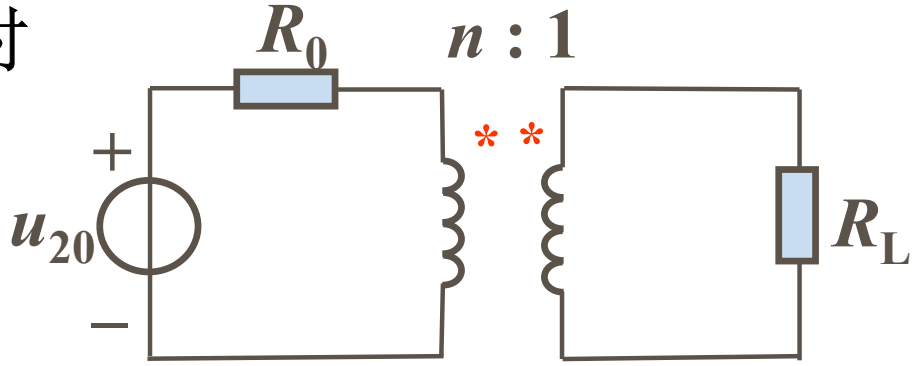
开路电压 $\dot{U}_{20}=\mathrm{R}_1\cdot(-\dot{I}_2)=2\times\dfrac{5}{2}\angle 0^\circ=5\angle 0^\circ$ V



等效阻抗 $\mathrm{R}_0=\dfrac{U_{20}}{-I_2}=2\,\Omega$ $\left(-\dot{\mathrm{I}}_2=\dfrac{\dot{\mathrm{U}}_1}{\mathrm{r}}=\dfrac{10\angle 0^\circ}{4}=\dfrac{5}{2}\angle 0^\circ\ \mathrm{A}\right)$

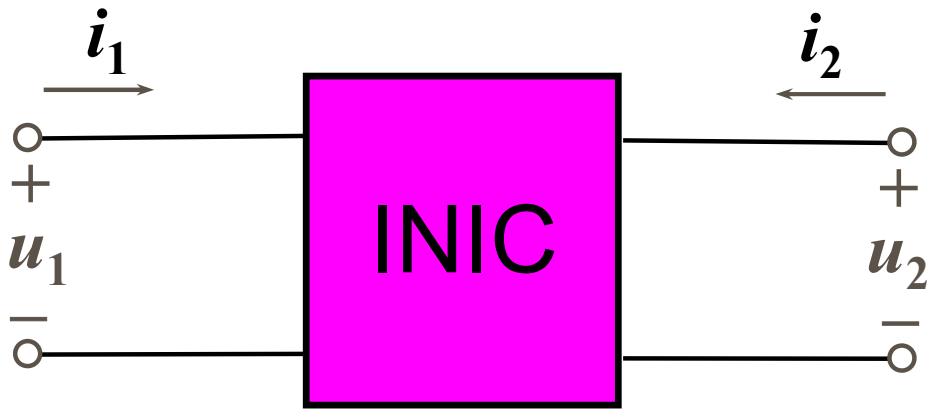
当 $\mathrm{n}^2\mathrm{R}_{\mathrm{L}}=\mathrm{R}_0$ $\mathrm{R}_{\mathrm{L}}=\dfrac{\mathrm{R}_0}{\mathrm{n}^2}=\dfrac{1}{2}\Omega$ 时

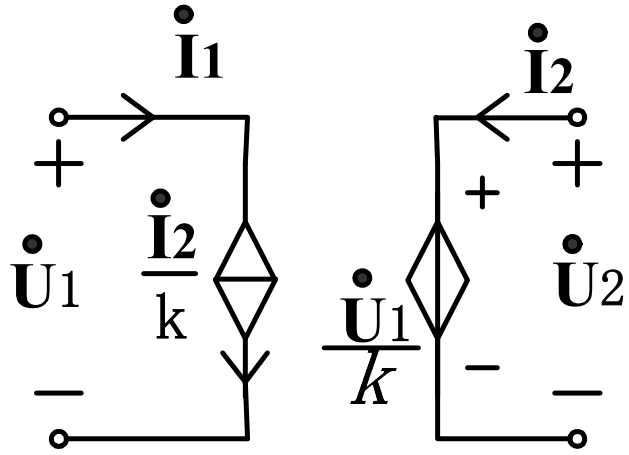
$$\mathrm{P}_{\max}=\frac{\mathrm{U}_{20}^2}{4\mathrm{R}_0}=\frac{25}{8}\omega$$



10.11 负阻抗变换器

1）负阻变换器





$$\begin{cases} u_1 = ku_2 \\ i_1 = \dfrac{1}{k} i_2 \end{cases}$$

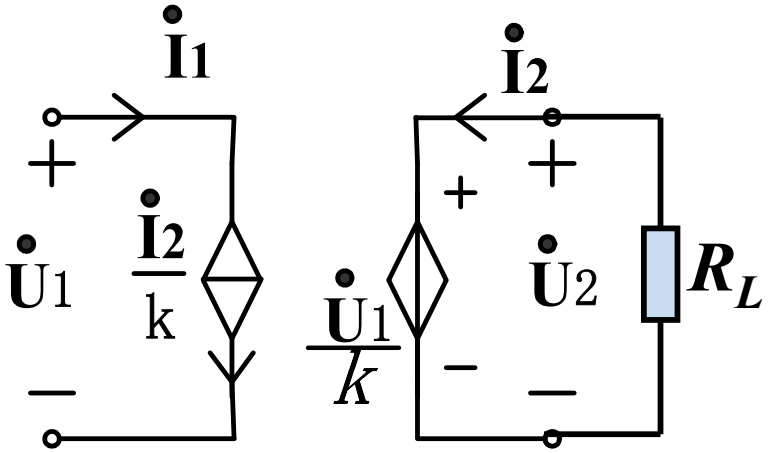
$$T = \begin{bmatrix} k & 0 \\ 0 & \dfrac{-1}{k} \end{bmatrix}$$

2) 负阻变换器应用

$2-2'$接电阻 R_L，$1-1'$ 看入等效阻抗

$$Z_\lambda = \frac{\dot{U}_1}{\dot{I}_1} = \frac{k\dot{U}_2}{\frac{1}{k}\dot{I}_2} = -k^2\frac{\dot{U}_2}{\left(-\dot{I}_2\right)} = -k^2 R_L$$

$1-1'$端等效为一个负电阻。



$$\dot{U}_1 = k\dot{U}_2$$

$$\dot{I}_1 = \frac{1}{k}\dot{I}_2$$

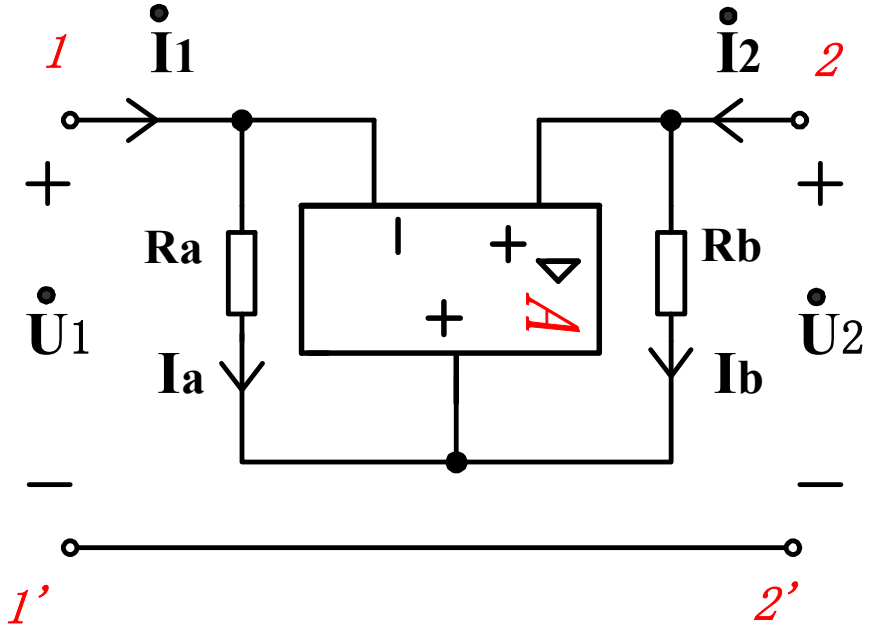
实现了负阻抗的变换！

应用：**RLC**串联揩振电路串入一负电阻**-R**后可产生自激正弦振荡波。

3)用单运放实现负阻变换器

$$\begin{cases} \dot{U}_1 \approx \dot{U}_2 \\ R_a \dot{I}_a - R_b \dot{I}_b = \dot{U}_1 - \dot{U}_2 = 0 \end{cases}$$

$$\dot{I}_1 \approx \dot{I}_a = \frac{R_b}{R_a}\dot{I}_b \approx \frac{R_b}{R_a}\dot{I}_2$$

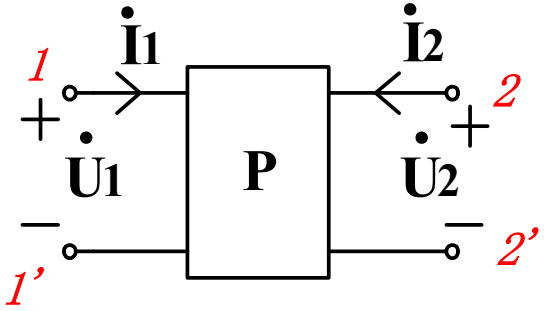


单运放组成的负阻变换器

$$R' = \frac{\dot{U}_1}{\dot{I}_1} = \frac{\dot{U}_2}{\frac{R_b}{R_a}\dot{I}_2} = -\frac{R_a}{R_b}\frac{\dot{U}_2}{-\dot{I}_2} = -\frac{R_a}{R_b}R_L$$

本章小结：

1. 双口网络定义,参数计算及互换



$$\begin{cases}\dot{U}_1 = Z_{11}\dot{I}_1 + Z_{12}\dot{I}_2 \\ \dot{U}_2 = Z_{21}\dot{I}_1 + Z_{22}\dot{I}_2\end{cases} \qquad \begin{cases}\dot{I}_1 = Y_{11}\dot{U}_1 + Y_{12}\dot{U}_2 \\ \dot{I}_2 = Y_{21}\dot{U}_1 + Y_{22}\dot{U}_2\end{cases} \qquad \begin{cases}\dot{U}_1 = A\dot{U}_2 + B(-\dot{I}_2) \\ \dot{I}_1 = C\dot{U}_2 + D(-\dot{I}_2)\end{cases}$$

互易：$Z_{12}=Z_{21}$；　$Y_{12}=Y_{21}$；　AD-BC=1。

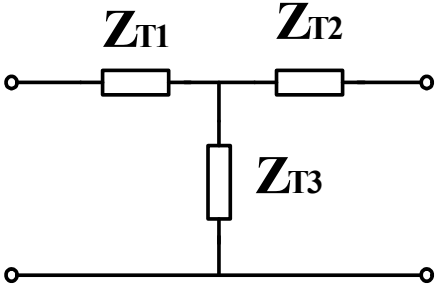
对称：$Z_{12}=Z_{21}, Z_{11}=Z_{22}$；　$Y_{12}=Y_{21}, Y_{11}=Y_{22}$；　AD-BC=1,A=D.

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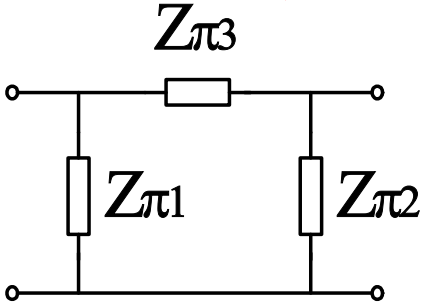
2. 双口网络等效电路

$$Z_{T1} = \frac{A-1}{C} \qquad Z_{T2} = \frac{D-1}{C} \qquad Z_{T3} = \frac{1}{C}$$

$$\begin{cases}Z_{T1} = Z_{11} - Z_{12} \\ Z_{T2} = Z_{22} - Z_{12} \\ Z_{T3} = Z_{12}\end{cases}$$



“T”型电路



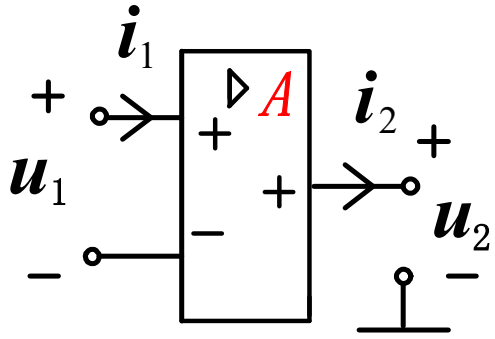
“π”型电路

$$Z_{\pi 1} = \frac{1}{Y_{11}+Y_{12}} \qquad Z_{\pi 2} = \frac{1}{Y_{22}+Y_{12}} \qquad Z_{\pi 3} = -\frac{1}{Y_{12}}$$

3. 双口网络连接

级联 $[T]=[T_A][T_B]$　并联 $[Y]=[Y_a]+[Y_b]$

串联 $[Z]=[Z_A]+[Z_B]$



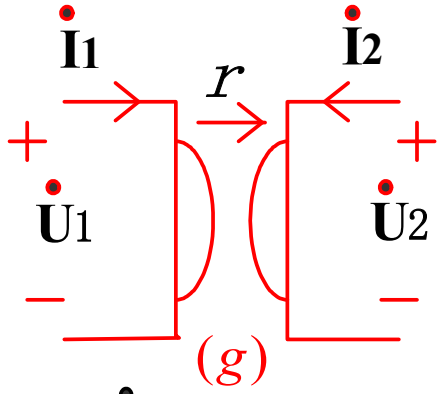
4. 运算放大器

“虚短”与“虚断”的概念；反相比例放大器；

同相比例放大器；　电压跟随器.

5. 回转器

$$\begin{cases}\dot{U}_1=-r\dot{I}_2\\ \dot{I}_1=\dfrac{1}{r}\dot{U}_2\end{cases}\qquad T=\begin{bmatrix}0 & r\\ \dfrac{1}{r} & 0\end{bmatrix}$$



6. 负阻变换器

$$\begin{cases}\boldsymbol{u}_1=k\boldsymbol{u}_2\\ \boldsymbol{i}_1=\dfrac{1}{k}\boldsymbol{i}_2\end{cases}\qquad T=\begin{bmatrix}k & 0\\ 0 & \dfrac{-1}{k}\end{bmatrix}$$

